

E.E. 3333-001: P01

Remote Event Sensing

4/24/08

Kyle Romero

Will Gray

Aaron Adcock

Ronjan Mathur

Adviser: Dr. Mitra

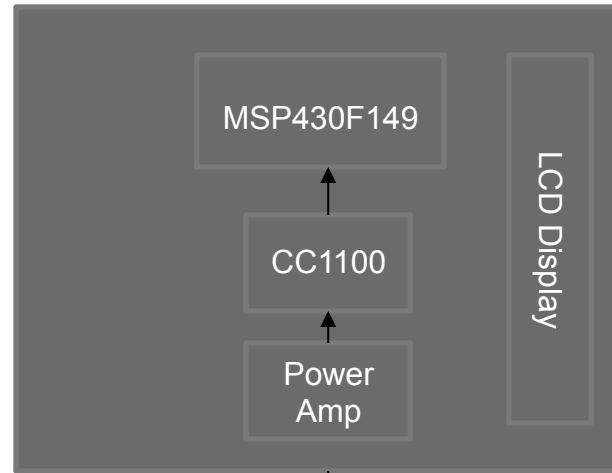
The Project

- The system should have two remote transmitters and one central receiver.
- An event occurs at transmitter 1.
- Within 15 minutes a corresponding event occurs at transmitter 2.
- The time between the two events is to be measured and must be accurate in seconds.
- A power amplifier should be used to maximize transmission distance

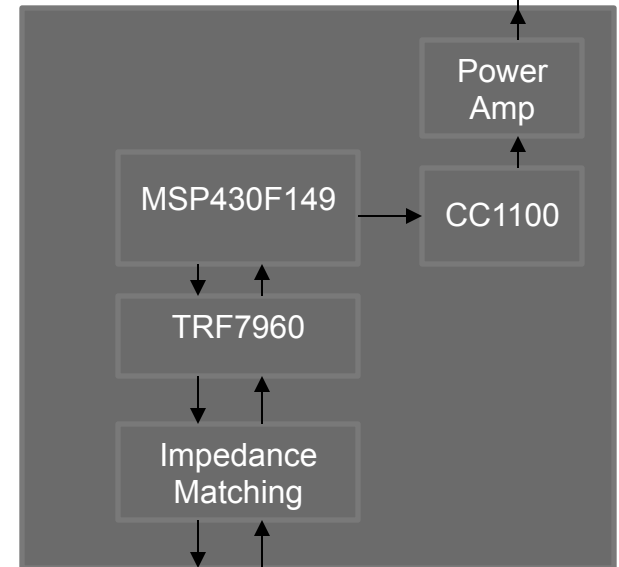
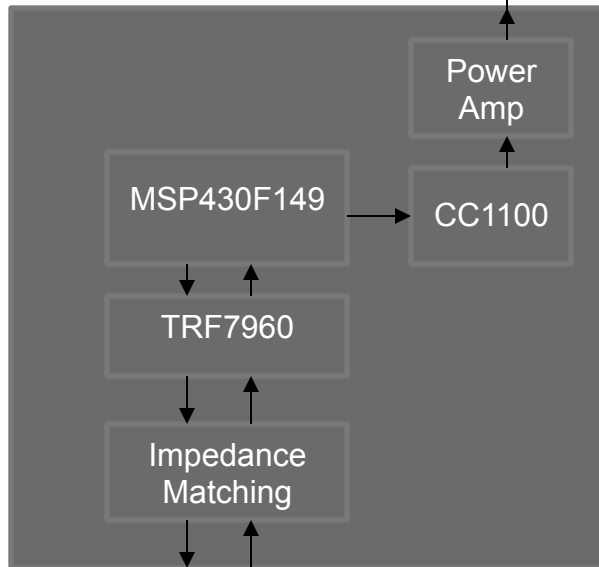
Project Breakdown

- Will Gray
 - CC1100/MSP430 Board
 - Receiver Board
- Aaron Adcock
 - Transmitter Board
 - RFID
- Kyle Romero
 - Software
- Ronjan Mathur
 - Power Amplifier

Receiver Board



Transmitter Boards



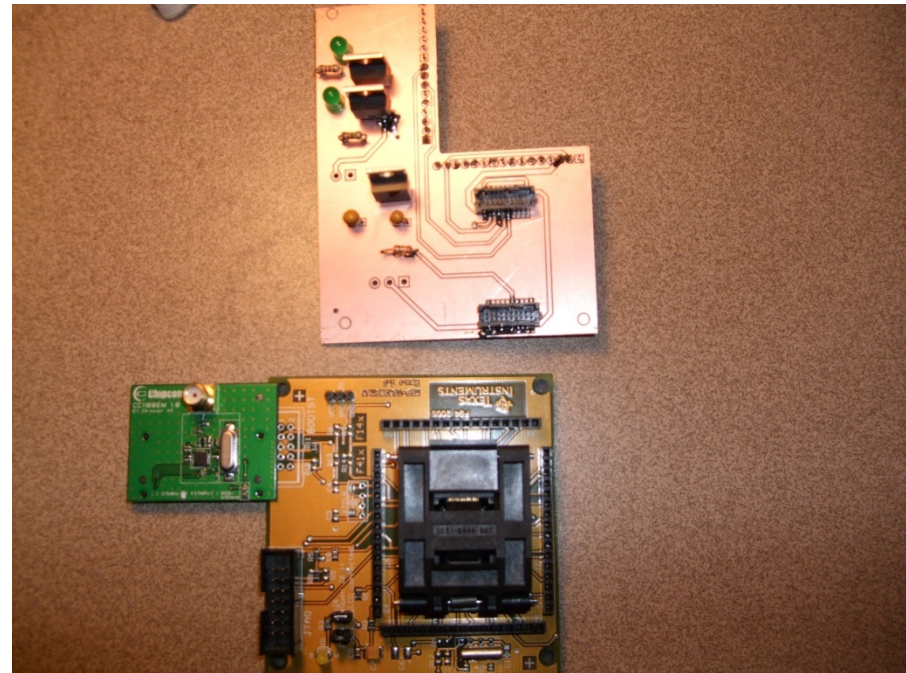
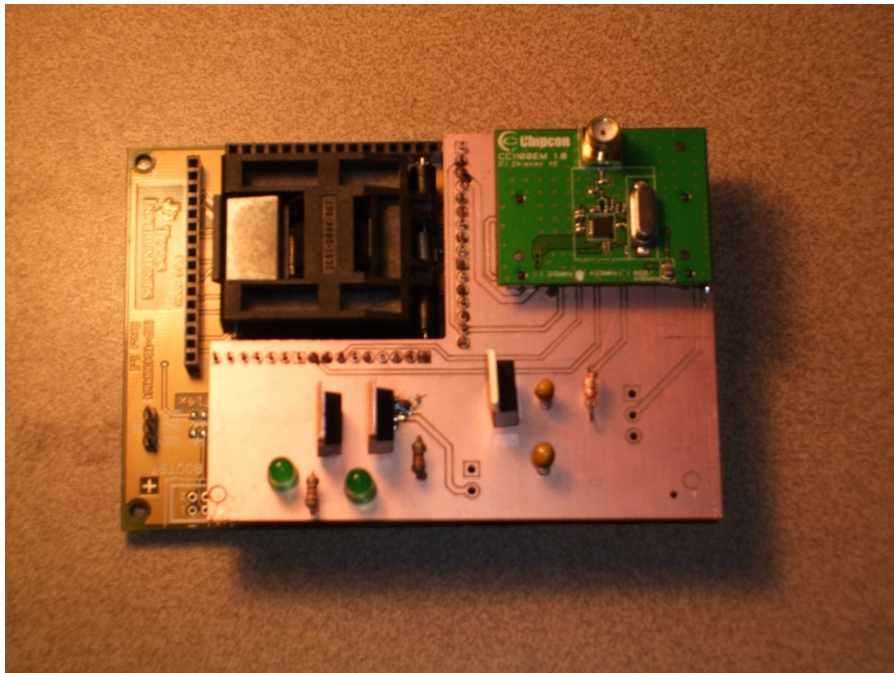
Event
Aaron Adcock

Event
RFID Antenna

CC1100/MSP430 Board

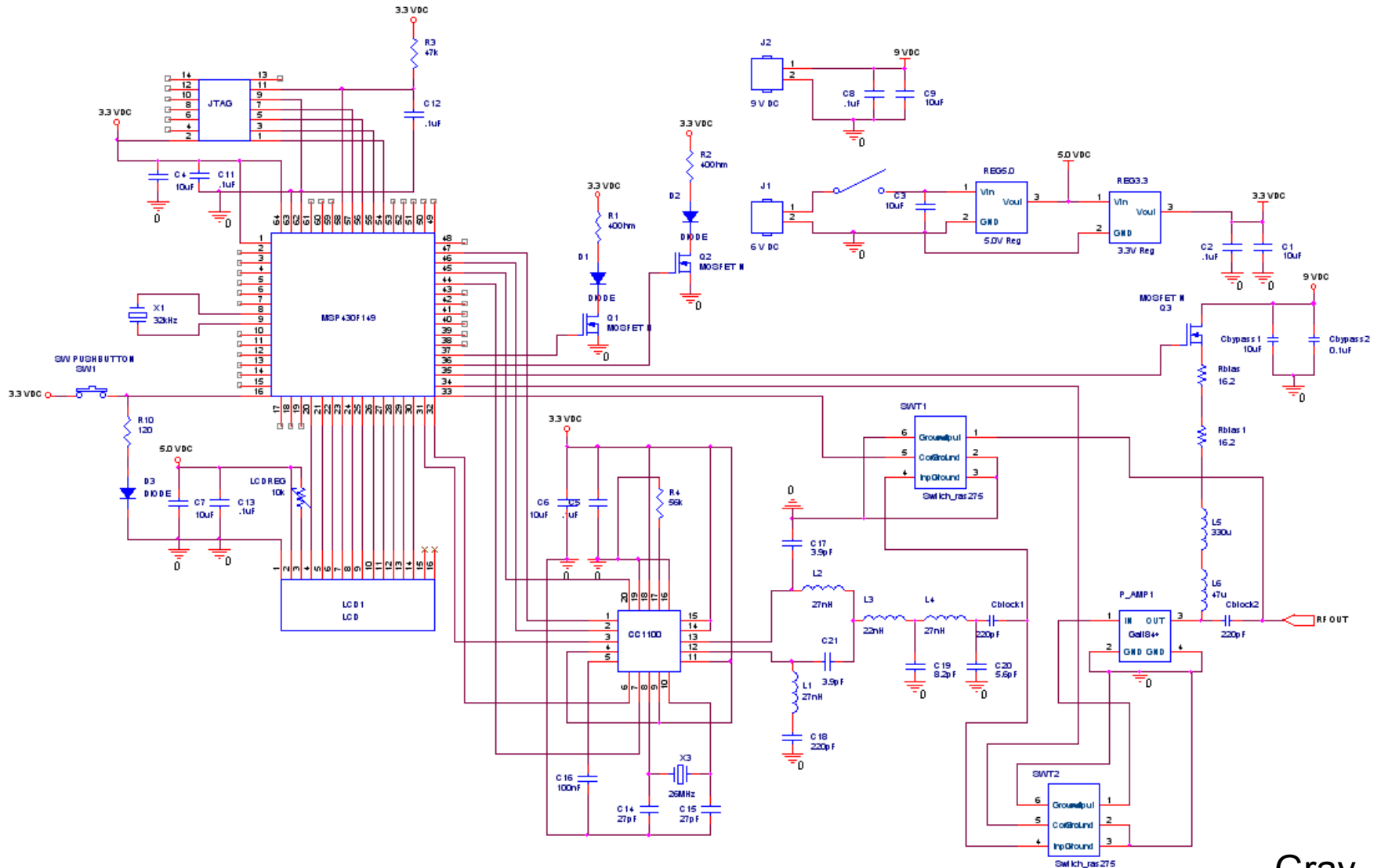
CC1100-MSP430 Board

- For Communication Between MSP430 Clamshell and CC1100 Daughterboard.



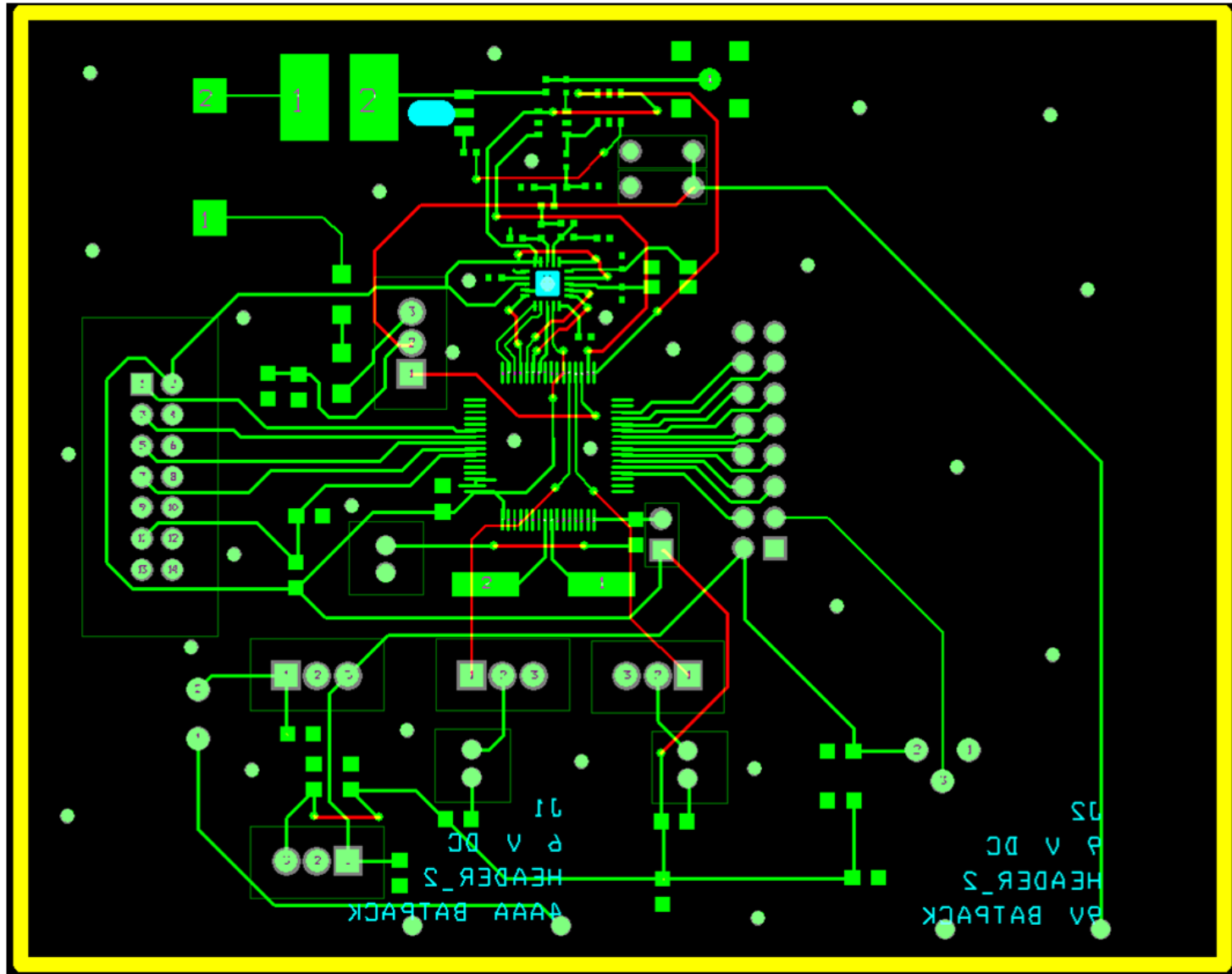
Receiver Board

Receiver Board

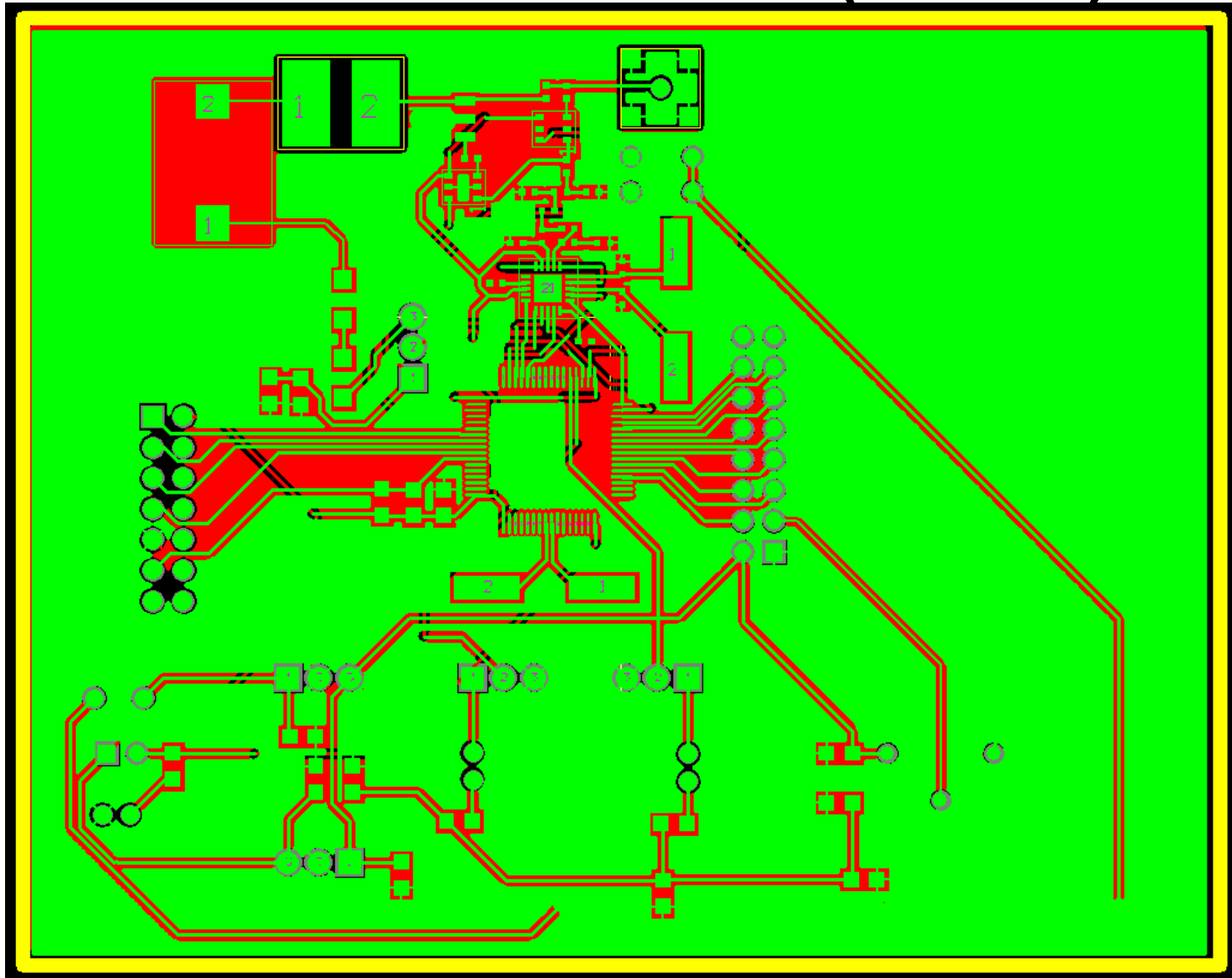


Gray

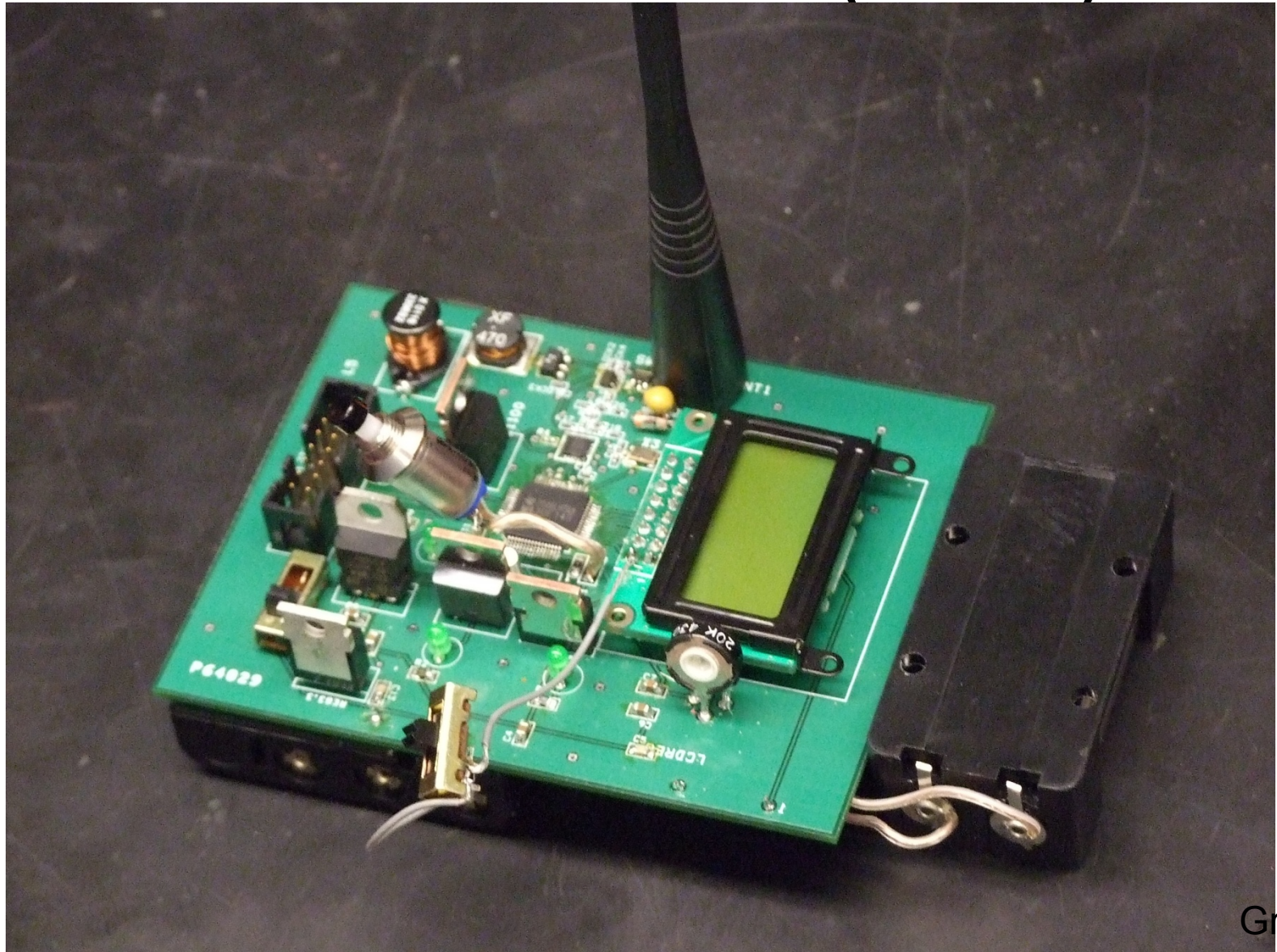
Receiver Board (cont.)



Receiver Board (cont.)



Receiver Board (cont.)



General Board Parts

General Board Parts Purchases			
Item	Qty.	CPU(Cost/Unit)	Total
MSP430F149	3	\$10.10	\$30.30
4 'AAA' Battery Mount	5	\$0.99	\$4.95
9V Batteries	5	\$3.10	\$15.50
9V Battery Mount	5	\$1.10	\$5.50
3.3V Linear Regulator	5	\$0.33	\$1.65
5.0V Linear Regulator	5	\$0.45	\$2.25
CC1100	5	\$4.61	\$23.05
100nF 0402	5	\$0.11	\$0.55
27pF 0402	10	\$0.05	\$0.50
3.9pF 0402	10	\$0.05	\$0.50
8.2pF 0402	5	\$0.05	\$0.25
5.6pF 0402	5	\$0.05	\$0.25
220pF 0402	15	\$0.10	\$1.50
27nH 0402	15	\$0.09	\$1.35
22nH 0402	5	\$0.07	\$0.35

General Board Parts (cont.)

General Board Parts Purchases			
Item	Qty.	CPU(Cost/Unit)	Total
56k ohm 0402	5	\$0.02	\$0.10
16.2 ohm 0402	5	\$0.10	\$0.50
120 ohm 0805	5	\$0.05	\$0.25
40 ohm 0805	5	\$0.05	\$0.25
16.2 ohm 0805	5	\$0.10	\$0.50
120 ohm 0805	5	\$0.05	\$0.25
26 MHz XTAL	5	\$1.65	\$8.25
RF Switch	10	\$0.11	\$1.10
Power Amplifier	3	\$1.10	\$3.30
IRL540N MOSFET	10	\$0.12	\$1.20
330uH 0805	5	\$0.05	\$0.25
47uH 0805	5	\$0.06	\$0.30
0.1uF 0805	15	\$0.05	\$0.75
10uF 0805	15	\$0.07	\$1.05
Coaxial Antenna Jack	4	\$5.43	\$21.72
32 kHz XTAL	5	\$0.12	\$0.60
	Total		\$128.82

General Board Parts (cont.)

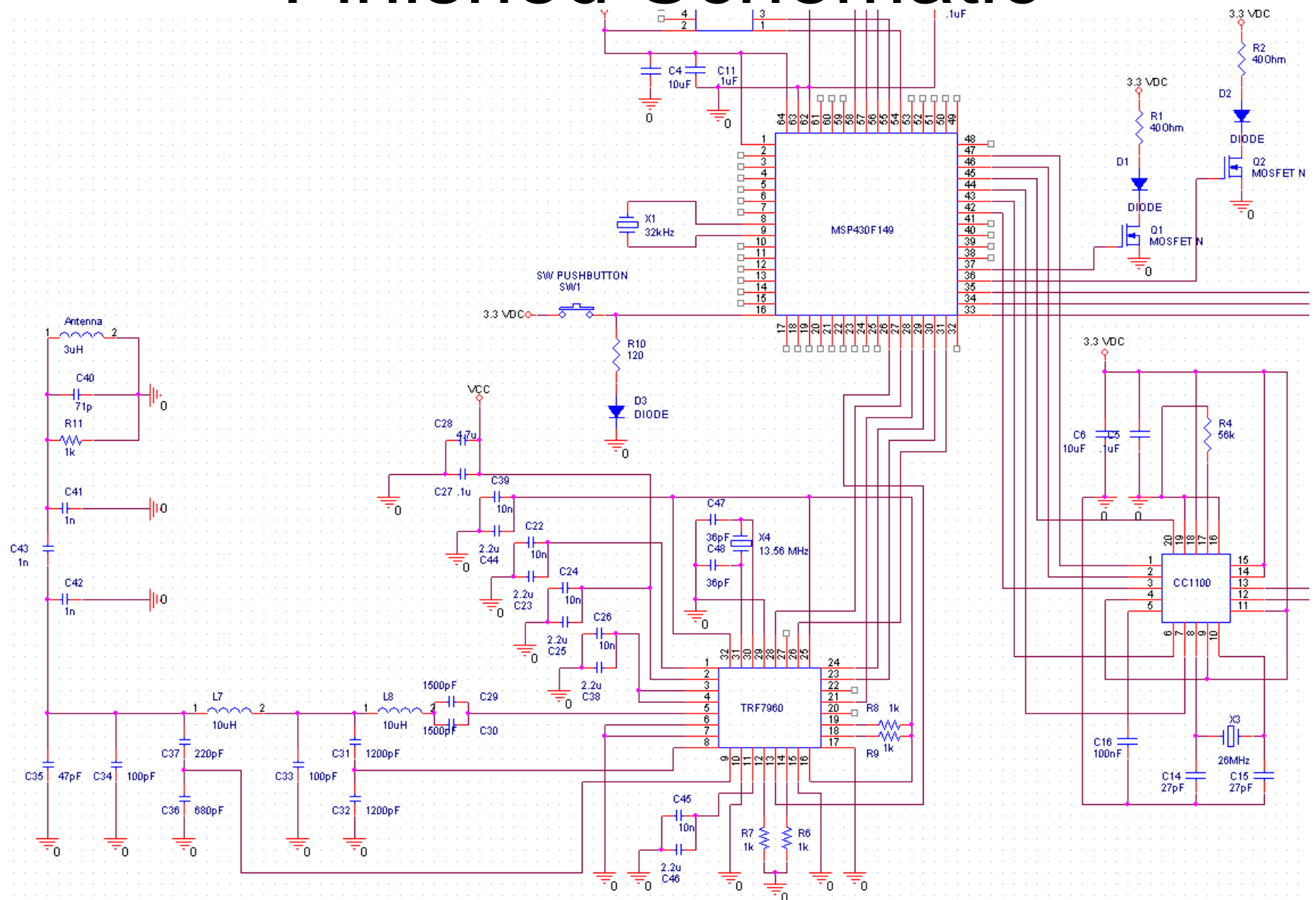
Receiver Board_0 Parts Purchases			
Item	Qty.	CPU(Cost/Unit)	Total
LEDs	8	\$0.12	\$0.96
JTAG Header	3	\$1.41	\$4.23
Pushbutton	2	\$4.45	\$8.90
AAA Batteries	9	\$0.99	\$8.91
	Total		\$23.00

Receiver Board Specific Parts

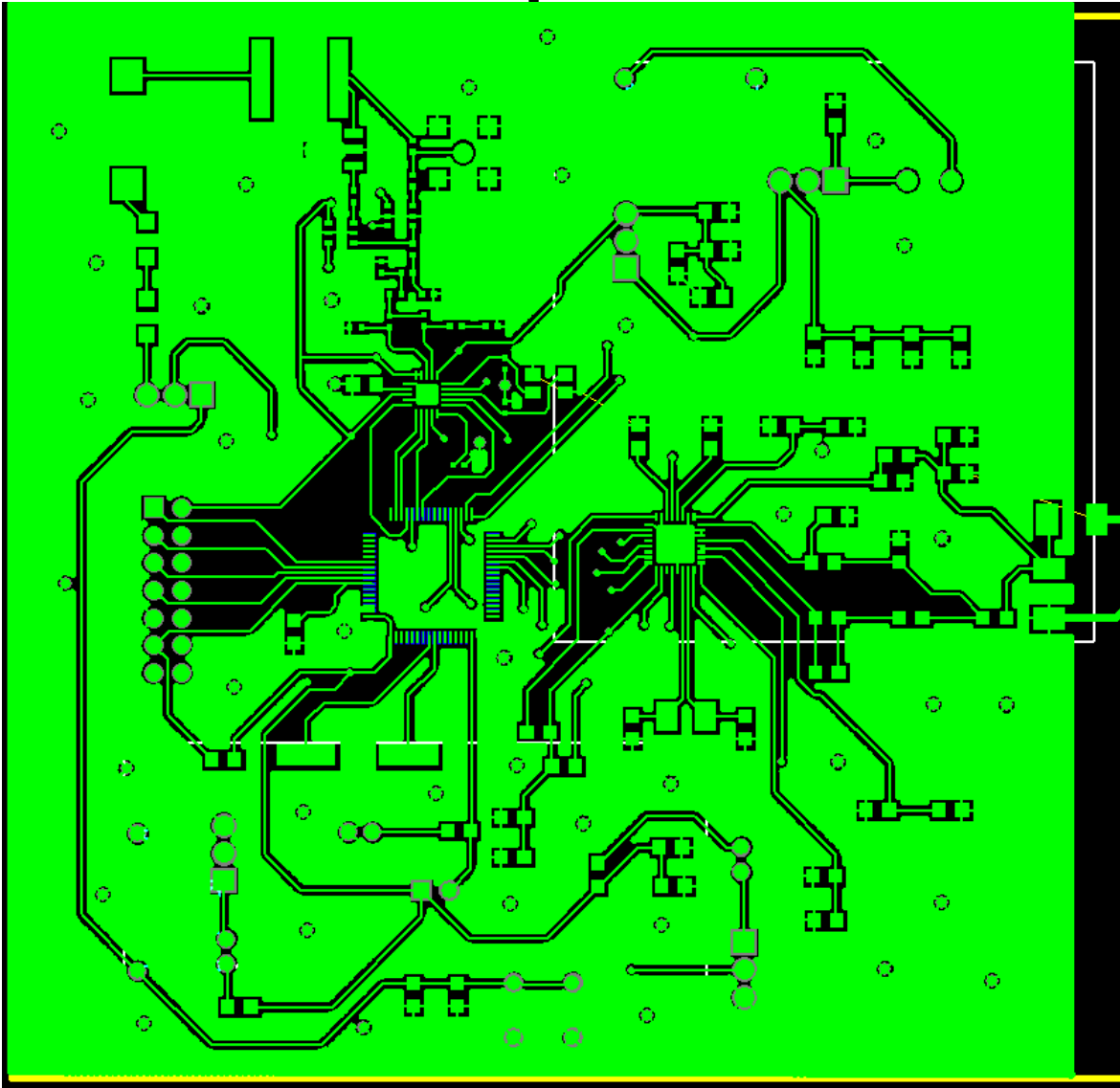
Receiver Board Parts Purchases			
Item	Qty.	CPU(Cost/Unit)	Total
0.1uF 0805	3	\$0.05	\$0.15
10uF 0805	3	\$0.07	\$0.21
10k Pot.	2	\$0.12	\$0.24
LCD Screen	1	\$22.19	\$22.19
	Total		\$22.79

Transmitter Board

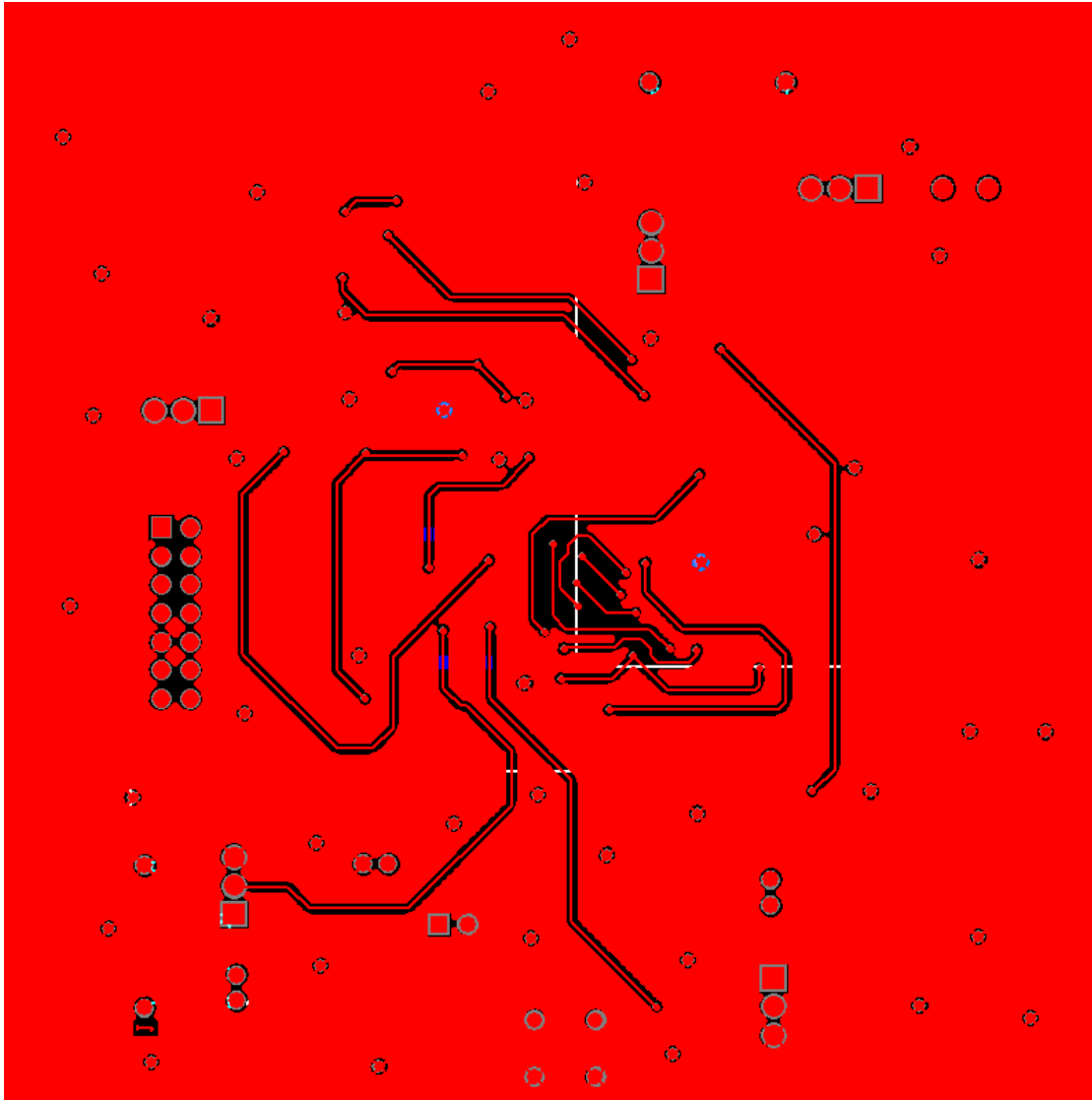
Finished Schematic



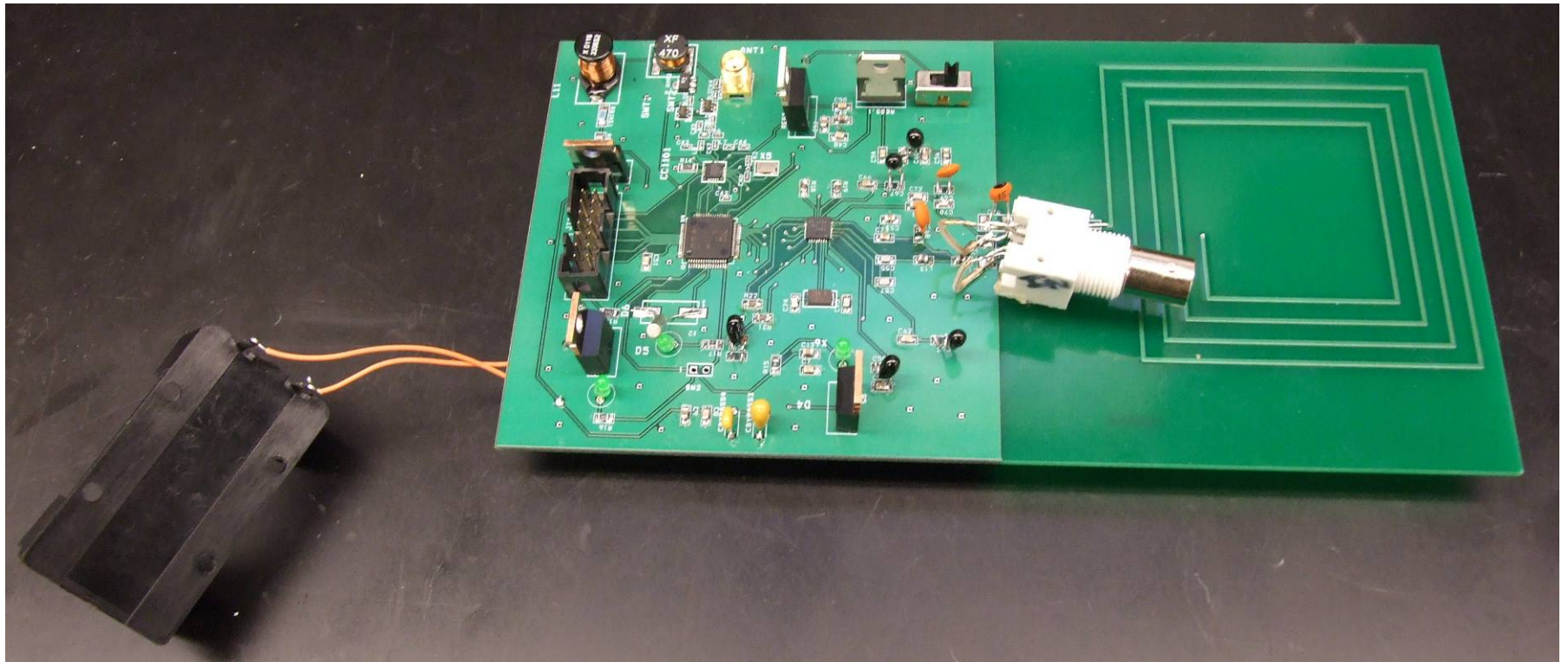
Top Side



Bottom Side



Populated Board



RFID

TRF7960

- 13.56 MHz
- Output power: 100, 200 mW (@5V)
- Flexible (various modulations/encoding options)
- Multiple protocols
- Made for use with TI MCU's
- Obtained for free

RFID Antenna

Square Spiral Inductor Formula

$$L = 9.375 \cdot \mu \cdot n^2 \cdot (d_{avg})^2 / (11 \cdot d_{out} - 7 \cdot d_{avg})$$

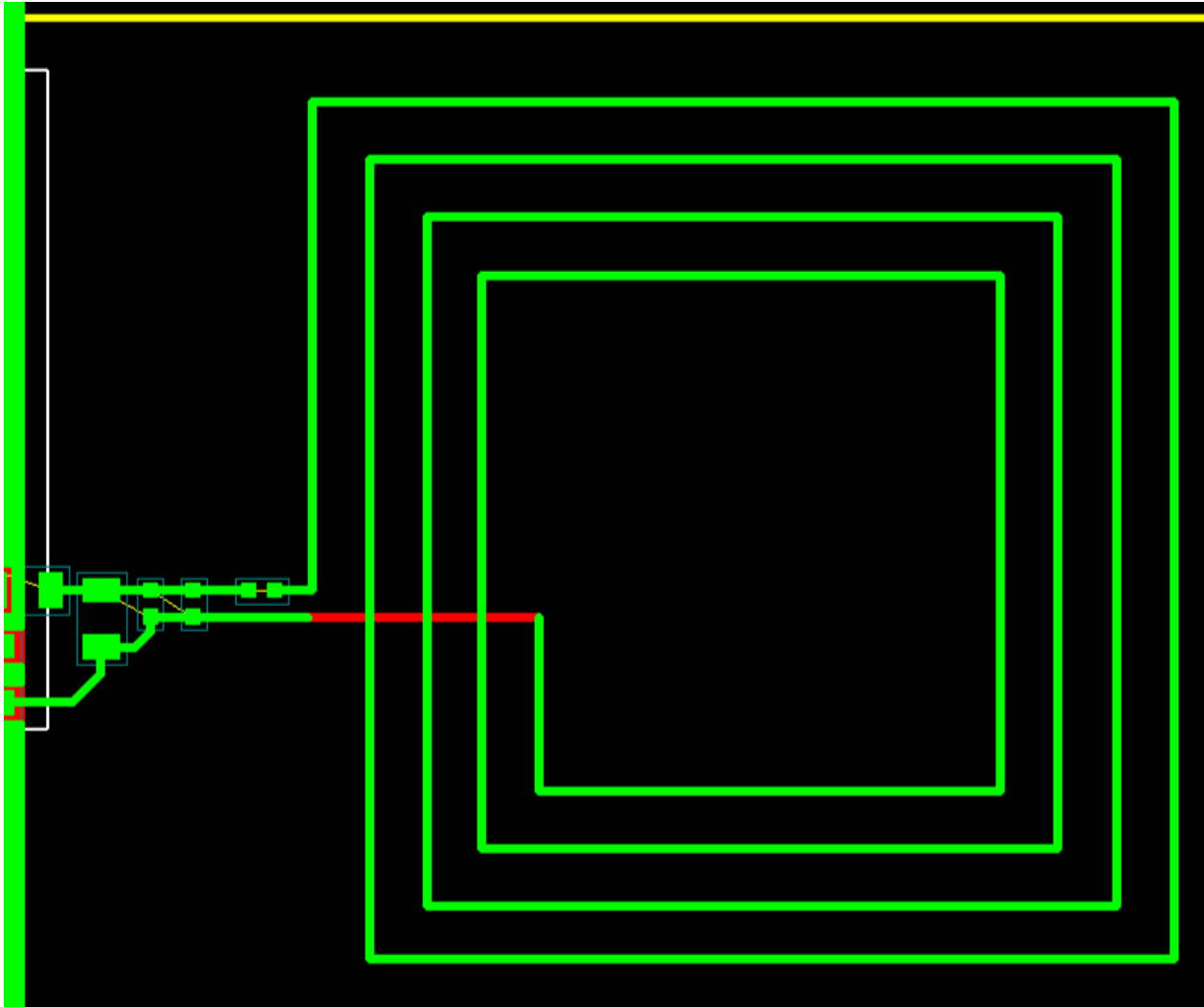
Specifically for Integrated Circuits

$$L = 1.94 \mu\text{H}$$

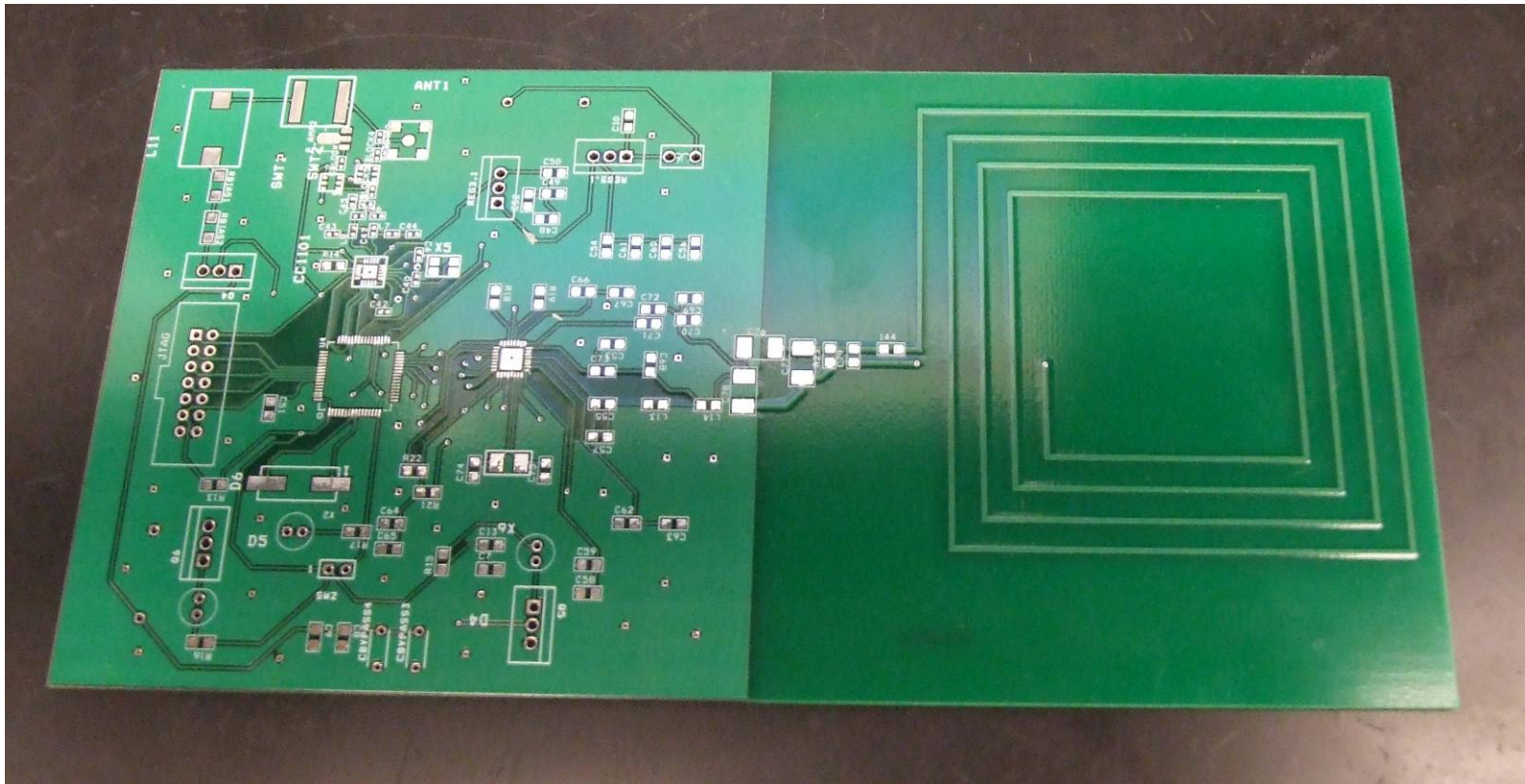
$$C = 71 \text{ pF}$$

Dr. Saed

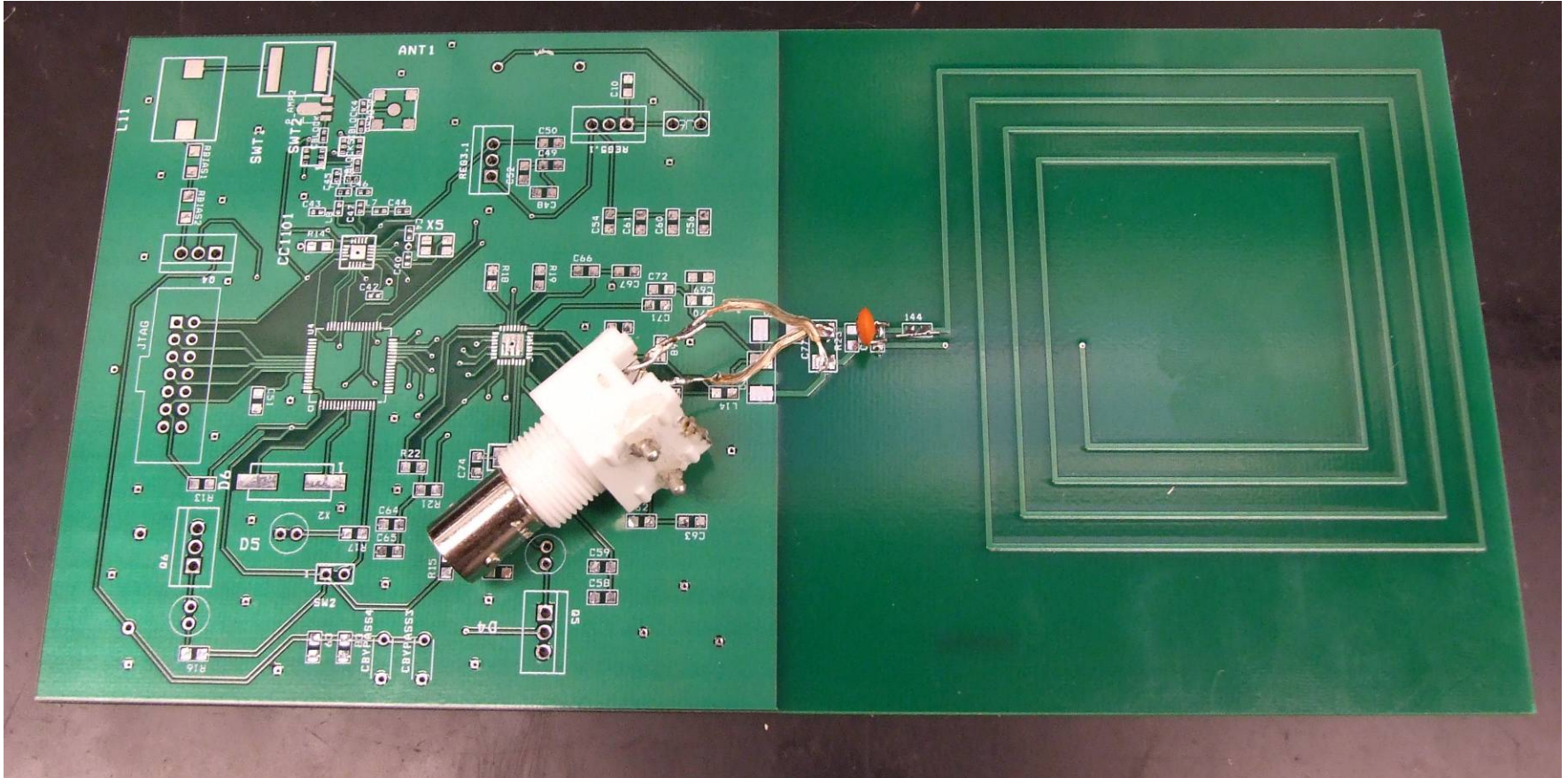
Antenna



PCB

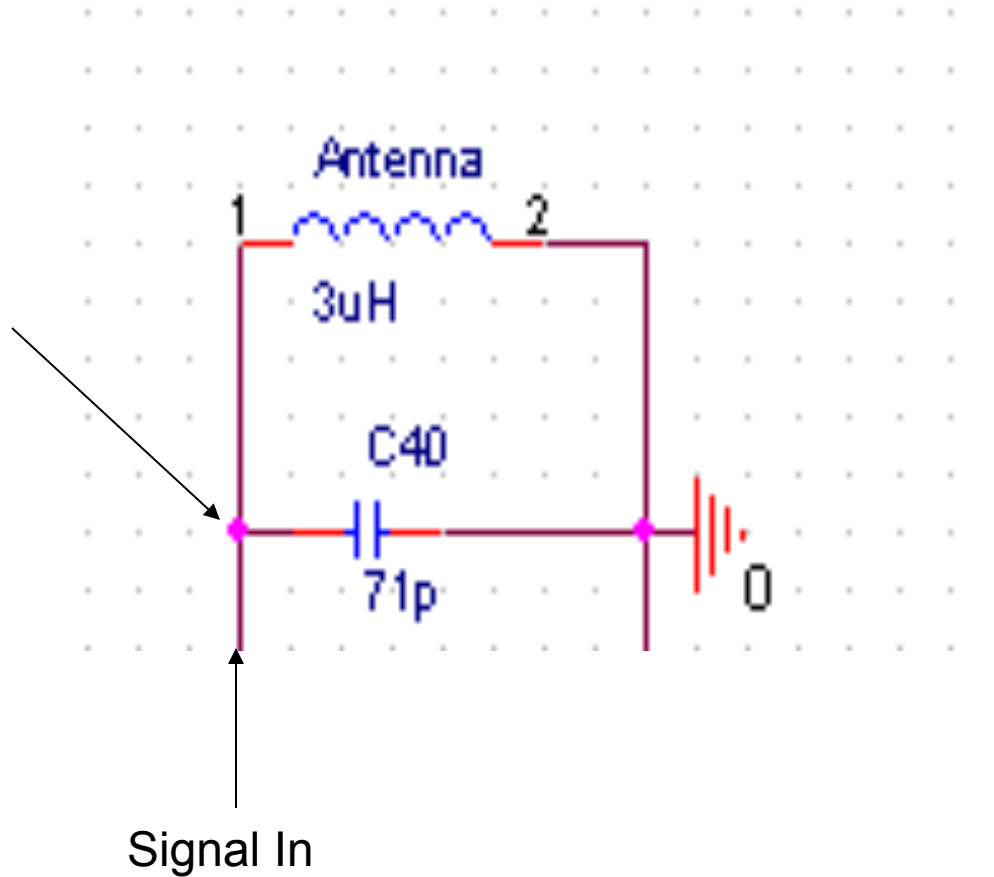


Antenna Testing

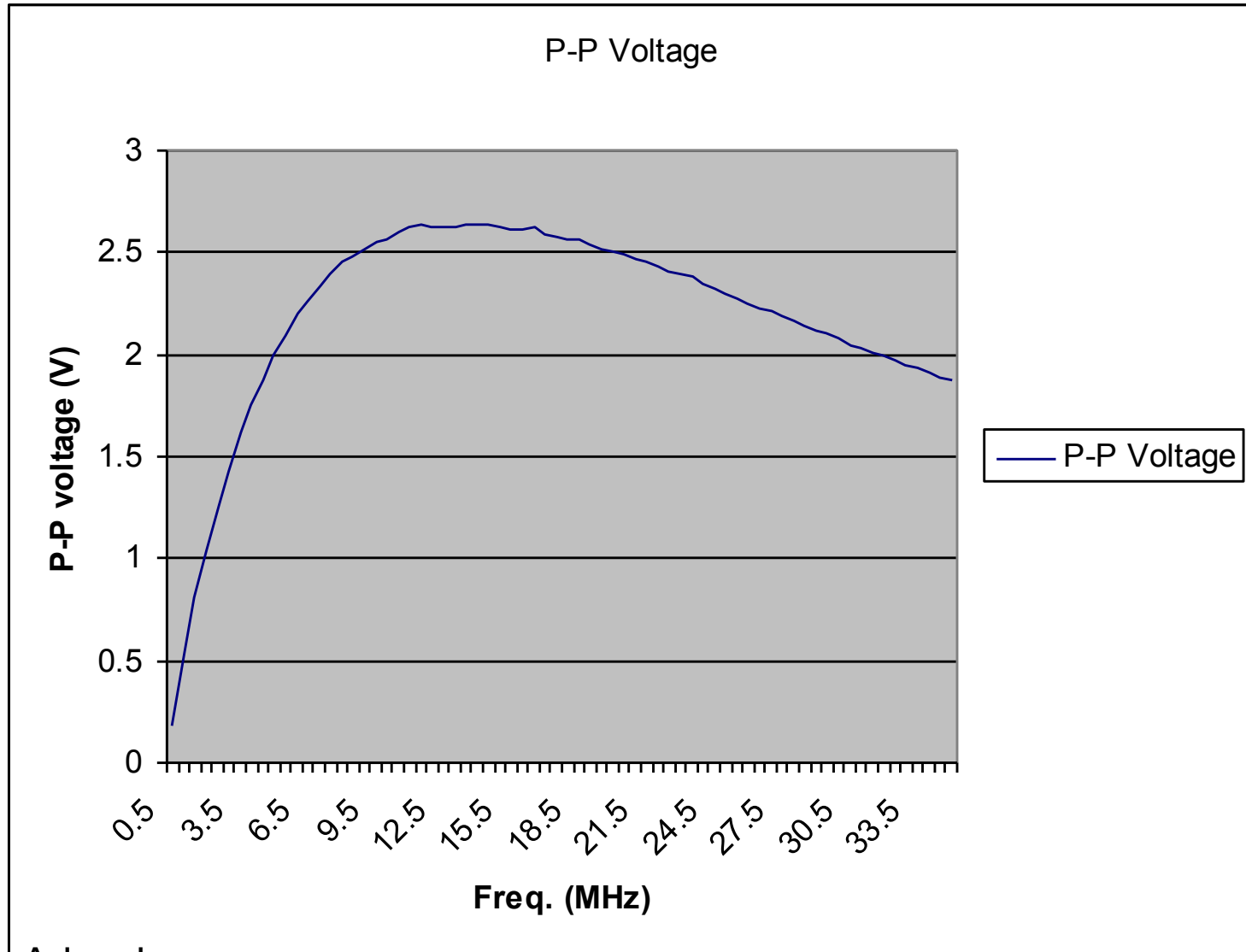


Test Circuit

10x Probe to
Oscilloscope



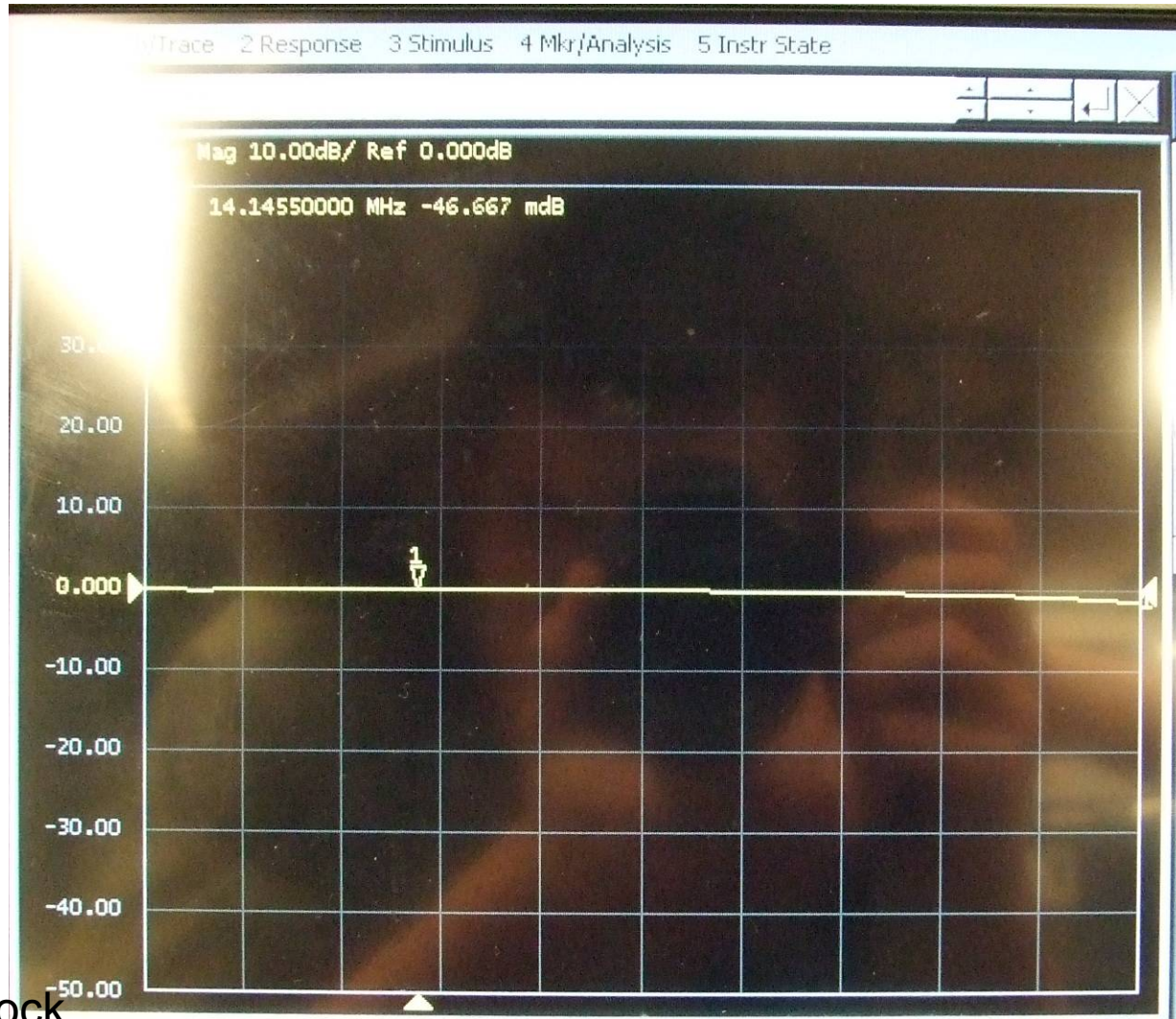
Resonance Data



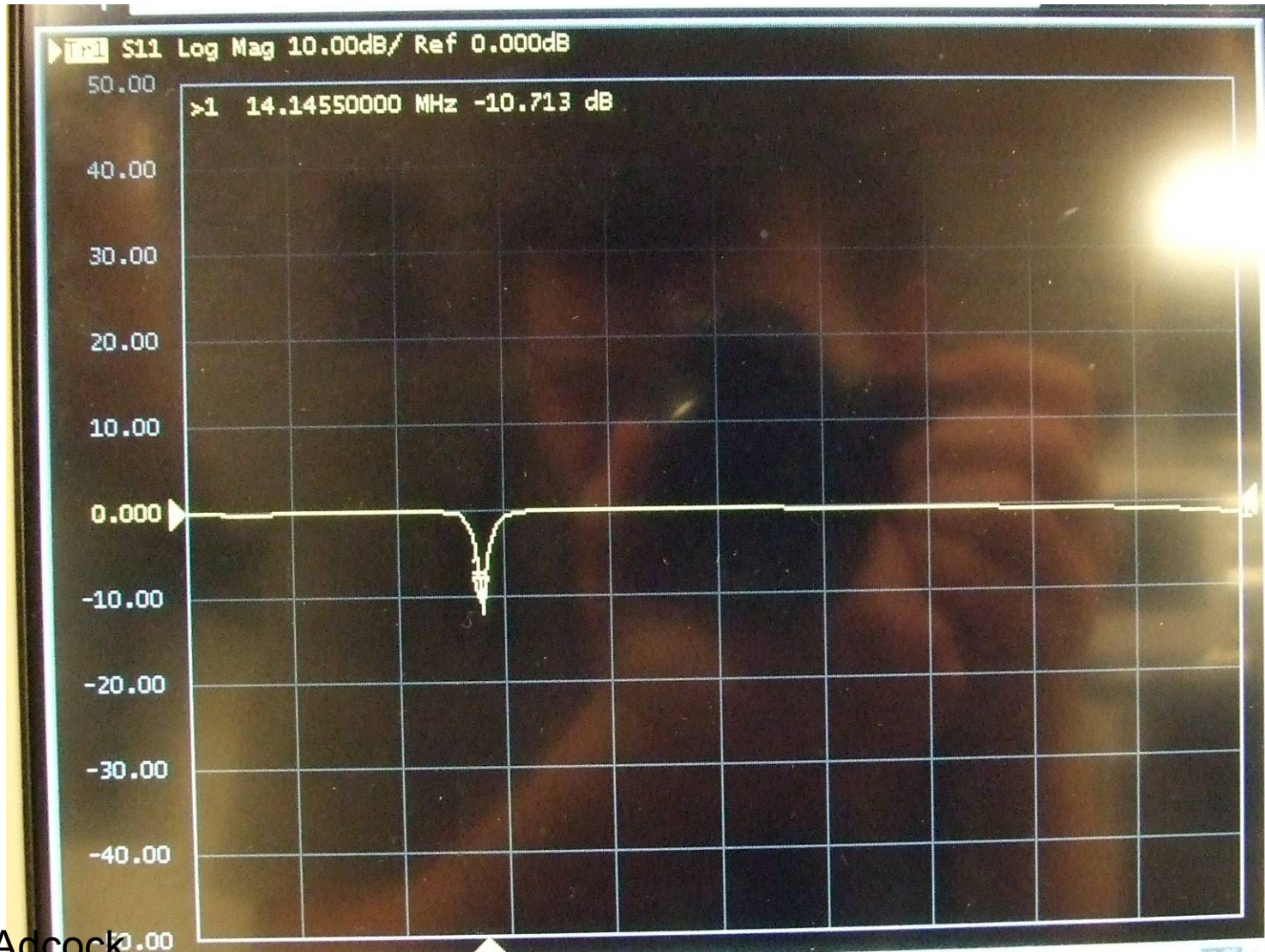
Testing Odd' s & Ends

- Using above data: 3-dB points at 4.5 MHz and 35 MHz
- $Q = \omega/BW = .648$
- Used Network Analyzer as well
- Attempted to create an impedance matching network

Network Analyzer



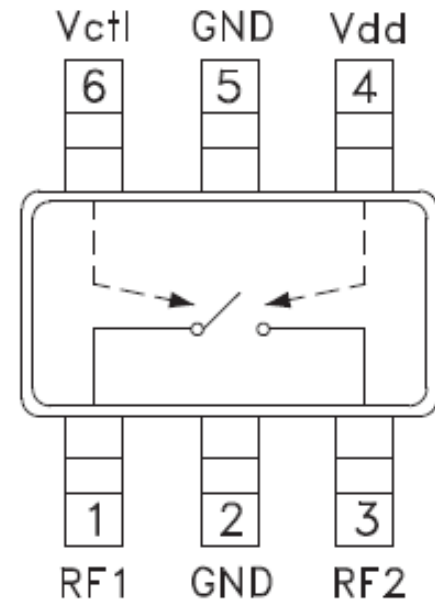
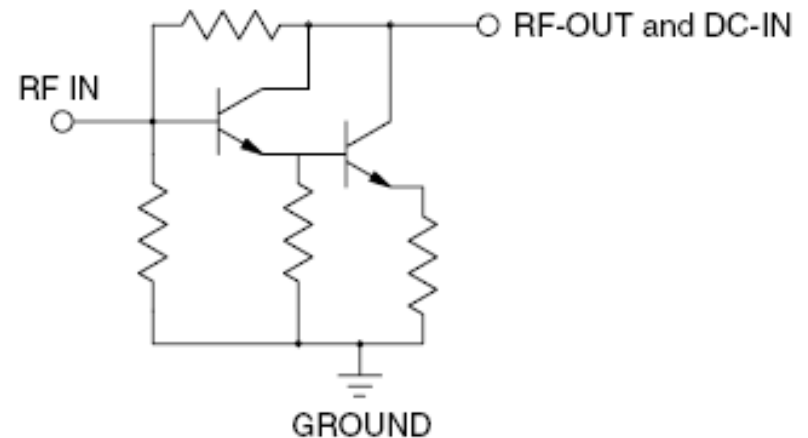
RFID Tag

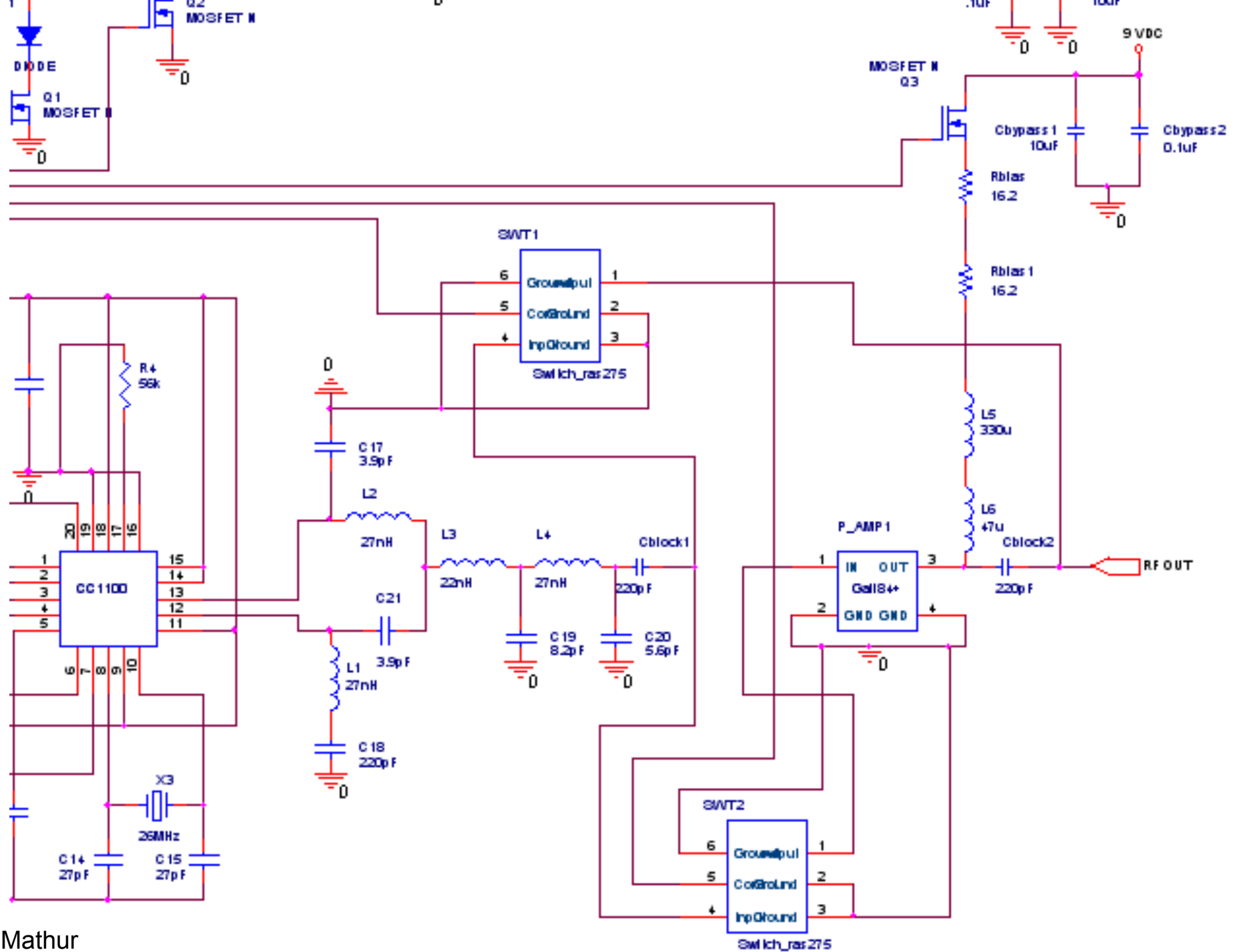


Power Amplifier

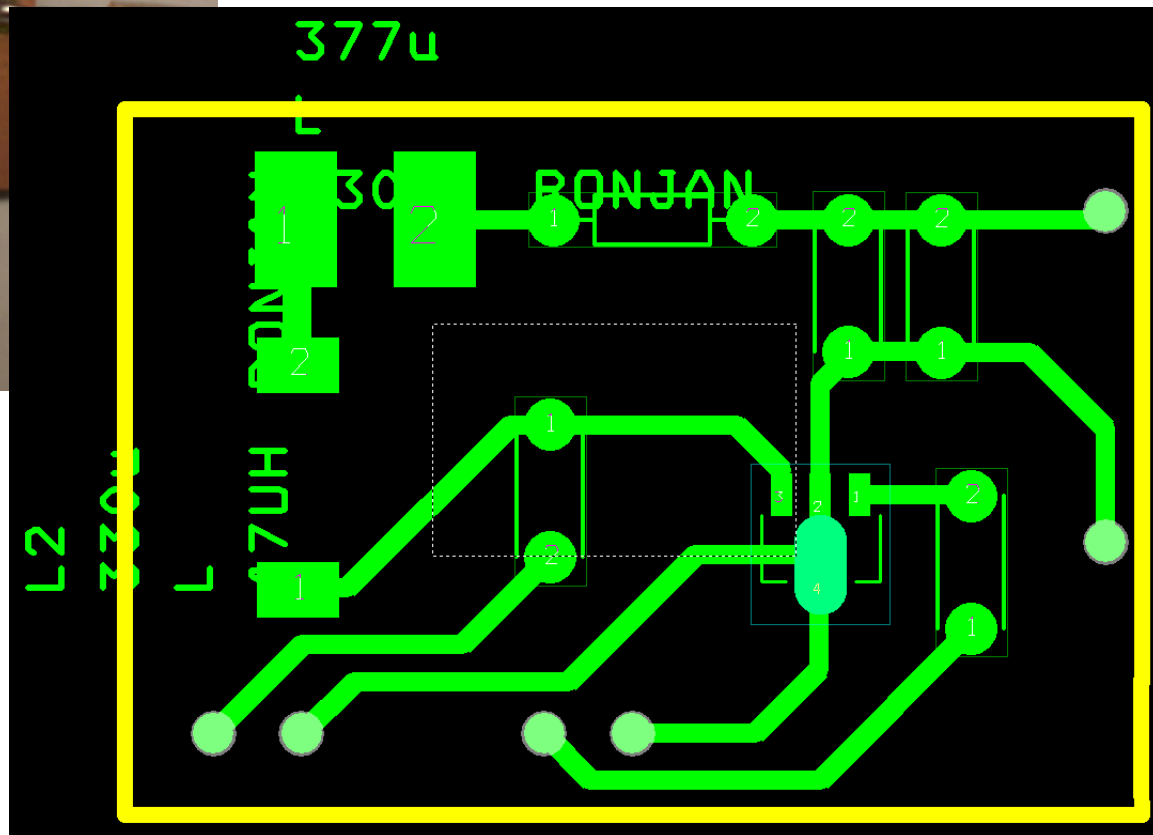
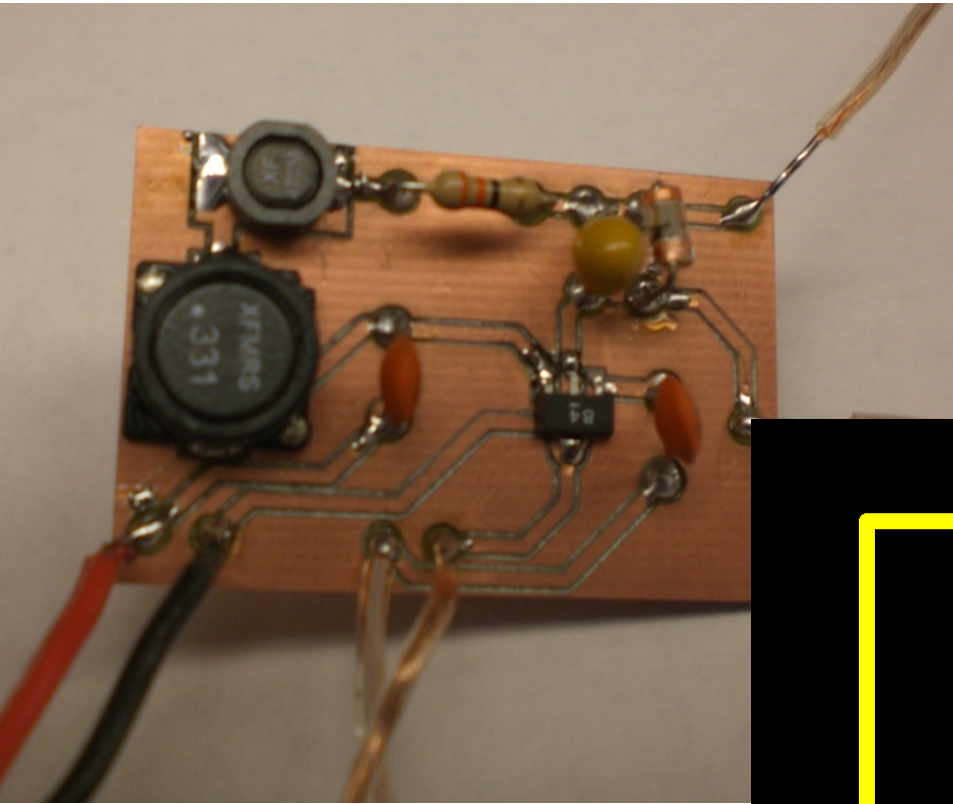
Power Amplifier

- Gali 84+ from MiniCircuits
 - Max Output @ 1-dB compression: 21.7 dBm
 - Input Gain: 21 dB
- Required to increase effective range of system
- Removable from circuit using HMC550 digitally controlled switch from Hittite
 - Low power loss switch, ~1 dBm
 - Quick Switching time
 - 25 dBm no-damage power rating

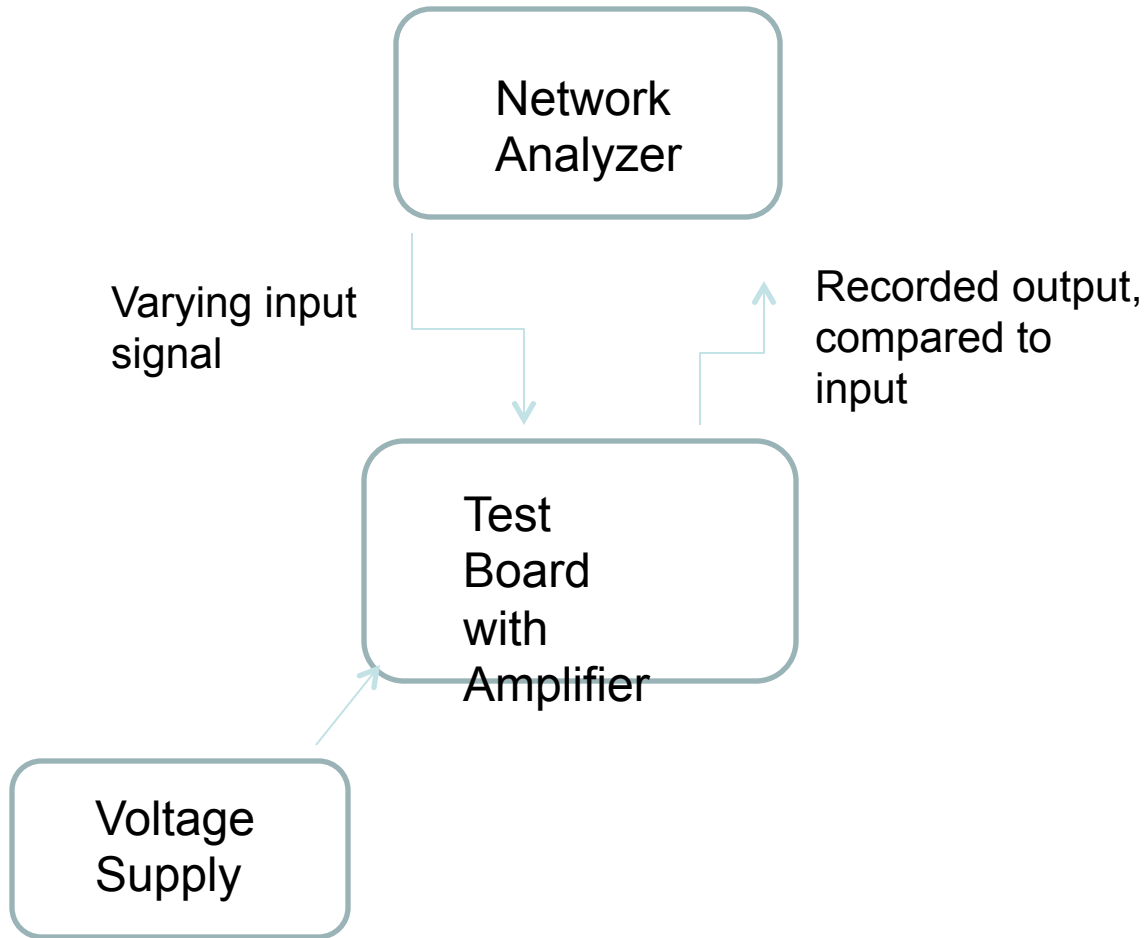




Amplifier Test Board



Amplifier Test



Amplifier Test



Components for RFID

Board

Part	Part #	Qty. (per board)	Cost per unit
2.2 uF Capacitor	81-GRM21R60J225KC15L	5	\$0.33
10 uF Capacitor	81-GRM21BR70J106KE76	5	\$0.48
4.7 uF Capacitor	81-GRM40034R475K6L	1	\$0.40
0.1 uF Capacitor	81-GRM40X104KA01D	1	\$0.08
1.2 uF Capacitor	80-C0805C125K9P	2	\$0.45
100 pF Capacitor	81-GRM40C101J50D	2	\$0.08
220 pF Capacitor	80-C0805C125K9P	1	\$0.06
680 pF Capacitor	80-C0805C681KDR	1	\$0.24
47 pF Capacitor	80-C0805C470J5G	1	\$0.05
1.5 nF Capacitor	81-GCM2195C2A152JA6D	2	\$0.16
36 pF Capacitor	80-C08055C360J5G	2	\$0.12
100 pF Capacitor	81-GRM40C101J50D	2	\$0.08

Parts List

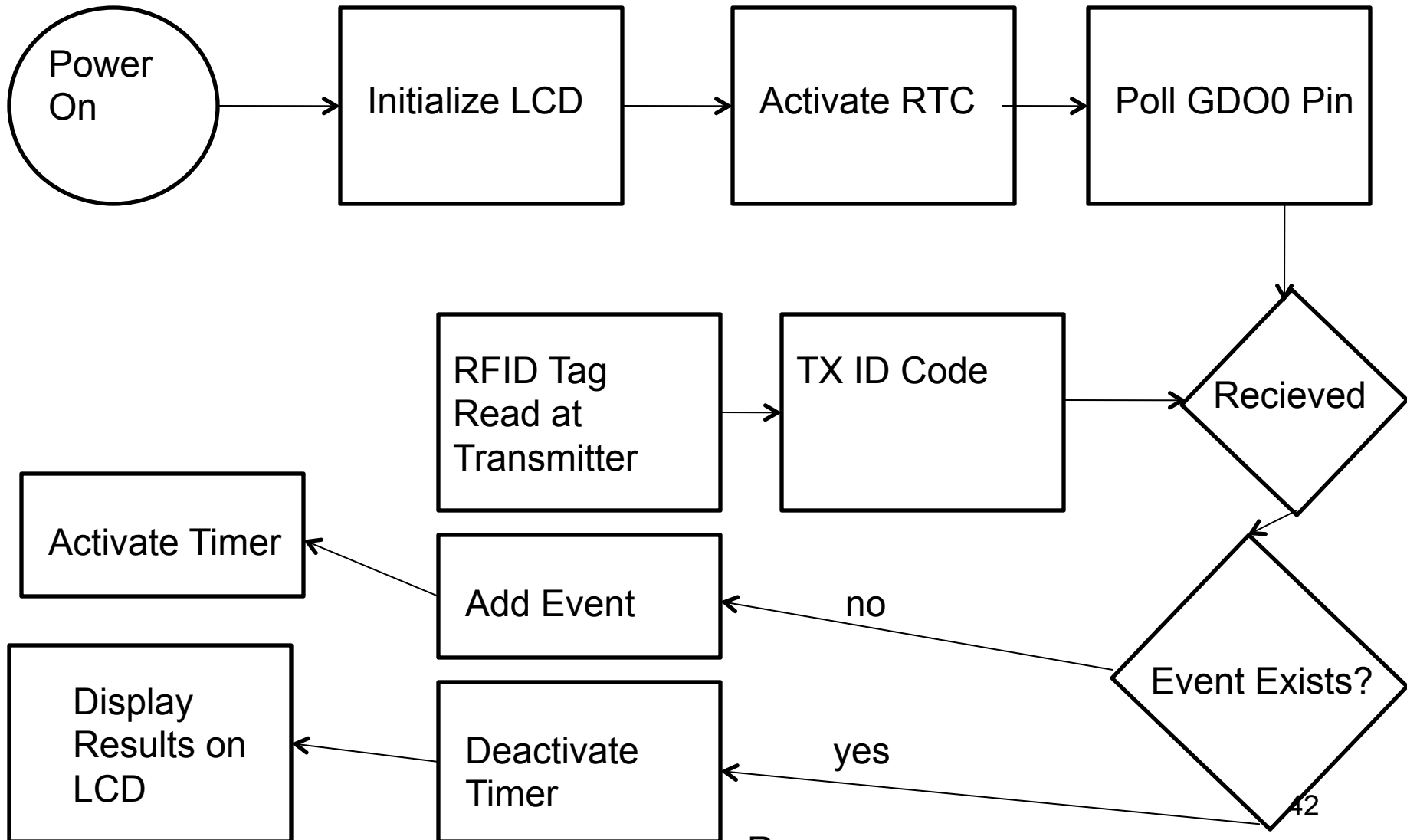
	part number	Amt. (per board)	Cost Per Unit (\$)
27 pF	GRM1555C1H270JZ01D	2	0.05
3.9 pF	GRM1555C1H3R9CZ01D	2	0.05
8.2 pF	GRM1555C1H8R2DZ01D	1	0.05
5.6 pF	GRM1555C1H5R6DZ01D	1	0.05
220 pF	140-CC505N221J	3	0.10
27 nH	LQG15HS27NJ02D	3	0.09
22 nH	MHI0402C22NJT-T	1	0.07
377uH	PE-53826S	1	1.31
16.2 Ω	RK73H2BLTD16R2F	2	0.10

Software

Overview

- Working:
 - LCD
 - Real Time Clock
- Coded but not tested
 - CC1100 interface
- Needs to be coded
 - RFID Interface

Software Flow Chart



P2.0	LCD	RS
P2.1	LCD	R/W
P2.2	LCD	R/W EN
P2.3	LCD	DB0
P2.4	LCD	DB1
P2.5	LCD	DB2
P2.6	LCD	DB3
P2.7	LCD	DB4
P3.0	LCD	DB5
P3.1	LCD	DB6
P3.2	LCD	DB7
P3.3	N/A	N/A
P3.4	N/A	N/A
P3.5	RFSW 1	P AMP ON/OFF
P3.6	RFSW 2	REG TX ON/OFF
P3.7	P AMP	Power ON/OFF
P4.0	POWER LED 1	LED 1 ON/OFF
P4.1	POWER LED 2	LED 2 ON/OFF
P4.2	N/A	N/A
P4.3	N/A	N/A
P4.4	N/A	N/A
P4.5	N/A	N/A
P4.6	CC1100	GDO2
P4.7	CC1100	GDO0
P5.0	CC1100	CSn
P5.1	CC1100	SI
P5.2	CC1100	SO/GDO1
P5.3	CC1100	SCLK

Important Pins

LCD

- Used a 2 Line x 8 Character Display from CrystalFontz
- 11 Pins are used
 - Rs: Enable Instruction or Data
 - R/W: Read or Write Mode
 - REN: Enable Pin
 - Push High then drop low to read data pins
 - DB0->DB7: Data Pins

Initialization

- Requires Predefined Commands to be sent at startup
- Care must be taken to avoid Delay errors or the display will not initialize correctly
- Once initialization is complete, the Cursor will appear at the top left corner of the LCD

upper 4 bits lower 4 bits	0 _a 0000 ₂	16 _a 0001 ₂	32 _a 0010 ₂	48 _a 0011 ₂	64 _a 0100 ₂	80 _a 0101 ₂	96 _a 0110 ₂	112 _a 0111 ₂	128 _a 1000 ₂	144 _a 1001 ₂	160 _a 1010 ₂	176 _a 1011 ₂	192 _a 1100 ₂	208 _a 1101 ₂	224 _a 1110 ₂	240 _a 1111 ₂
0 _a 0000 ₂	CGROM [0]			0aP`P									一タミxp			
1 _a 0001 ₂	CGROM [1]			!1AQa4									。アチ△äq			
2 _a 0010 ₂	CGROM [2]			"2BRbr									「イツ×pθ			
3 _a 0011 ₂	CGROM [3]			#3CScs									」ウテモε∞			
4 _a 0100 ₂	CGROM [4]			\$4DTdt									ゝイトμΩ			
5 _a 0101 ₂	CGROM [5]			%5EUeu									・オナ1cÜ			
6 _a 0110 ₂	CGROM [6]			&6FUfu									ヲカニヨpΣ			
7 _a 0111 ₂	CGROM [7]			*7GWgw									フキヌラgn			
8 _a 1000 ₂				(8HXhx									イウネリJΣ			
9 _a 1001 ₂				>9IYiy									ウケル'Y			
10 _a 1010 ₂				*:JZjz									エコハレj≠			
11 _a 1011 ₂				+;K[k<									オサヒロ*厶			
12 _a 1100 ₂				,<L¥1									カシフワΦ円			
13 _a 1101 ₂				-=M]m>									ユヌへンモ÷			
14 _a 1110 ₂				.>N^n→									ヨセホ*顔			
15 _a 1111 ₂				/?O_o←									ッソマ°ö■			

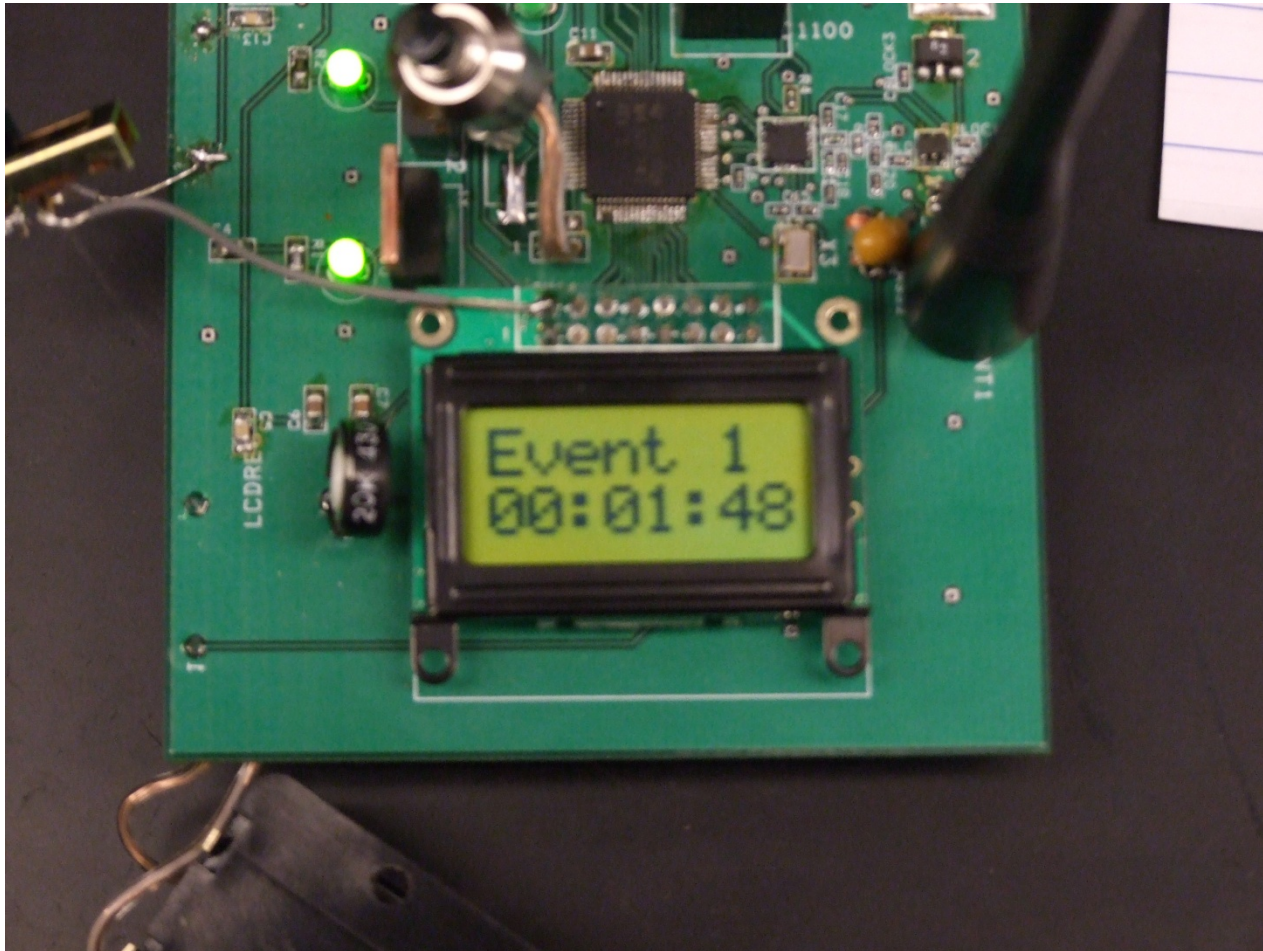
Character Map

Figure 8. Character Generator ROM (CGROM)

Sending Commands

```
void LcdWriteChar(unsigned char dval)
{
    P2OUT = LCD_RWEN+LCD_RS; // RS E
    P2OUT |= dval <<3;
    P3OUT = dval>>5;
    P3OUT+= POWERAMP;
    P2OUT ^= LCD_RWEN; //
    Delay(100); //waitUntilDone();
} // LcdWriteChar()
```

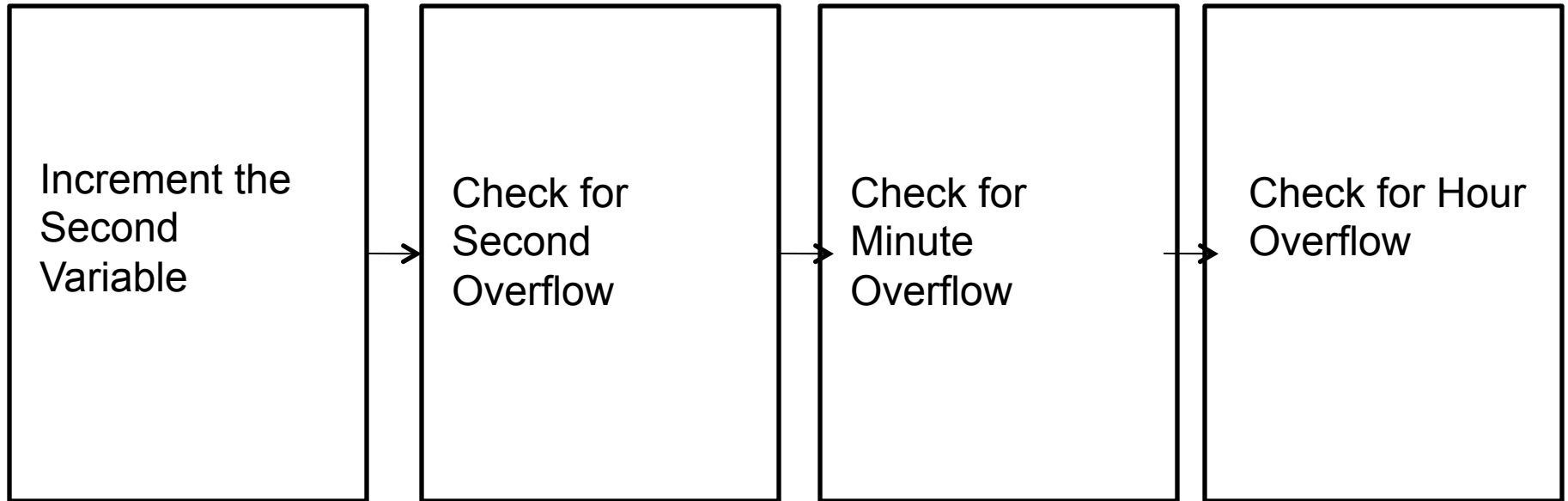
The LCD Working



The Timer

- Based off of a Timer A Interrupt
- Connected to the 32kHz Crystal
- Interrupt triggered on timer overflow
 - Set to overflow at 32000
 - This causes the interrupt to occur every 1s
- Upon interrupt enable, tick the real time clock and output current time onto LCD

Tick Command



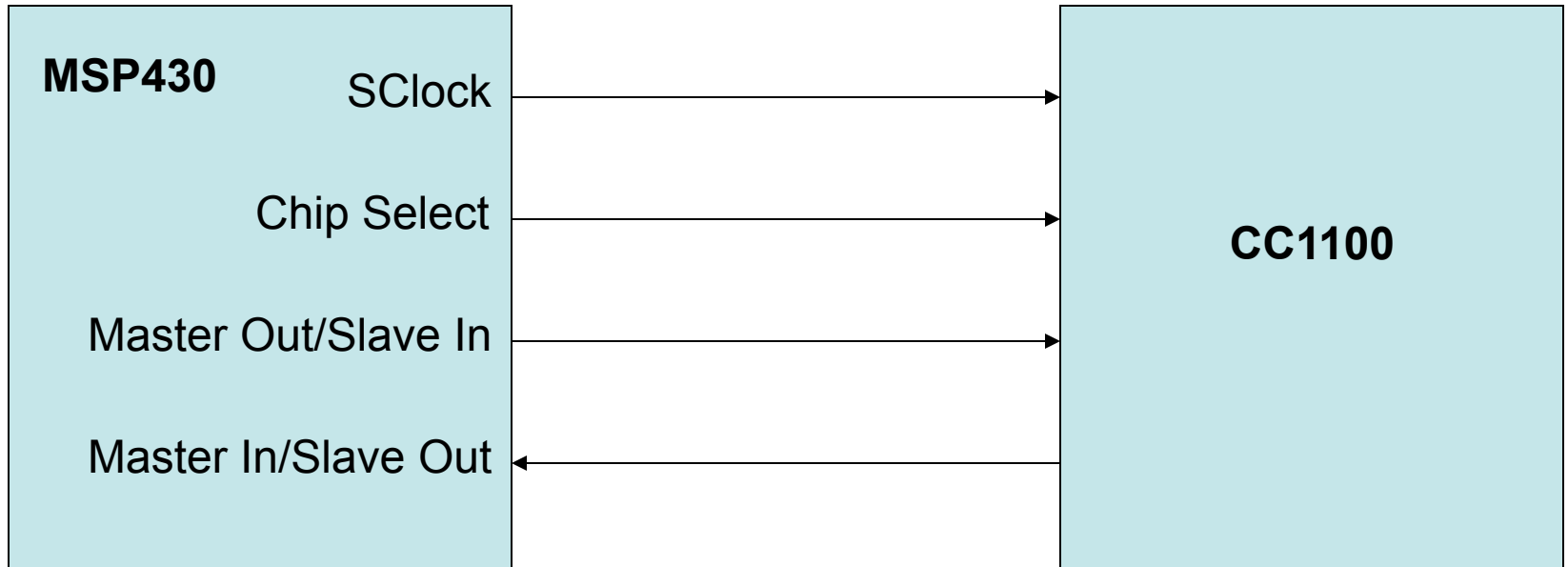
The CC1100

- GDO0 pin Connected to Port 4
- No interrupt pin on Port 4
 - Requires software polling to detect RX on GDO0 pin
- While Loop at end of main function that checks for signal and then retrieves the data from the transmission.

CC1100, continued

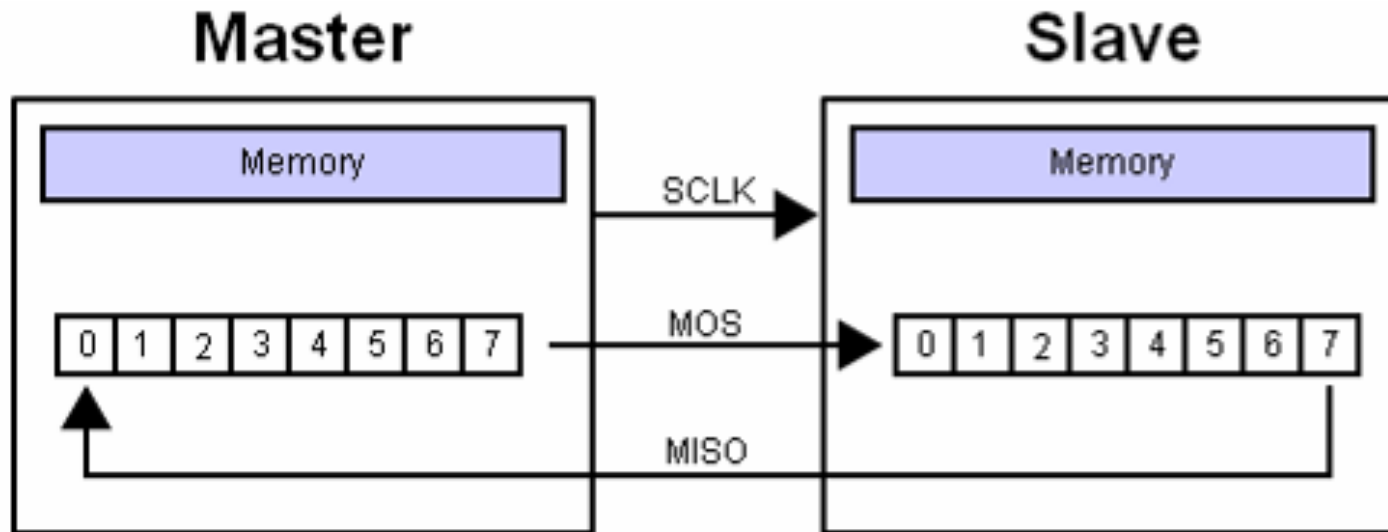
- Code still in progress
- Migrated Code from Eval Board Code
- Once completed, the transmitter board will transmit the ID read from an RFID tag and transmit it
- This will be the event ID and will be used to start and stop the appropriate event.

SPI Interface



SPI Operation

- For the CC1100
 - Lower Chip Select pin
 - Begin Transferring when CC1100 is ready
 - After transfer raise CS pin



http://en.wikipedia.org/wiki/Serial_Peripheral_Interface_Bus

Packets

- Blocks of digital information

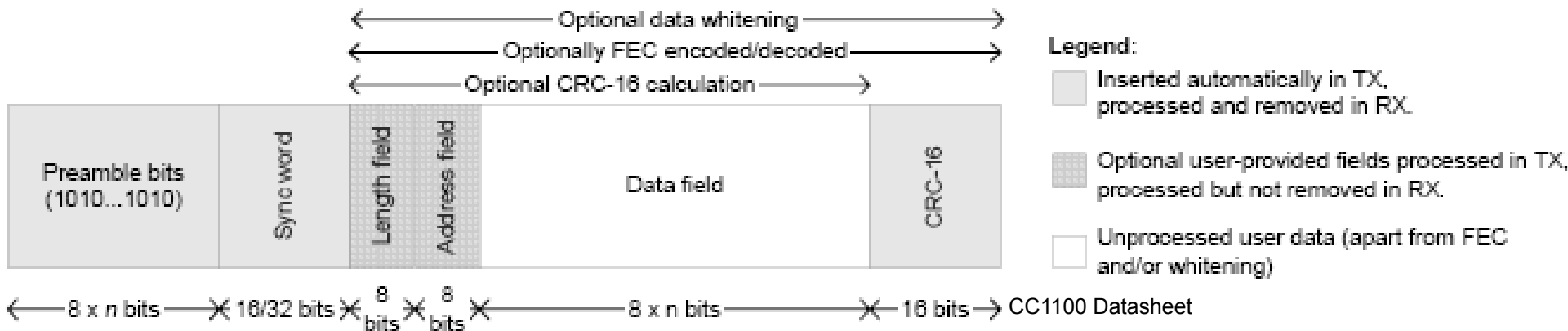


Figure 11: Packet Format

Project Status

Accomplishments

- Successful Transmission/Timing using Development Board
- Developed/Populated Receiver/Transmitter Boards
- Successful Programming of Both Boards
- Created/Tested Power Amplifier Circuit
- LCD Code works on new board
- Timing Code works on new board
- Created and tested a power amplifier circuit
- Designed/Built RFID Antenna

Project Status

To Be Completed:

- Communication between boards using CC1100
- Communication with RFID chip
- Successful detection/reading of RFID tag
- Successful transmission using power amplifier

Budget and Gantt Chart

Budget - Projected

Estimated Budget (1/18/2008)				
Labor Costs				
Name	Hourly Pay Rate	Hours Worked	Total Pay	
Ronjan Mathur	\$15.00	210	\$3,150.00	
Aaron Adcock	\$15.00	210	\$3,150.00	
Will Gray	\$15.00	210	\$3,150.00	
Kyle Romero	\$15.00	210	\$3,150.00	
		Total Labor Cost	\$12,600.00	
		75% Overhead	\$9,450.00	
		Total Labor and Overhead	\$22,050.00	
Parts Costs				
Part Name	Cost	Quantity	Total Cost	
Misc. Parts	\$50.00	NA	\$0.00	
		Total Parts Cost	\$0.00	
Equipment Rental Costs				
Equipment Name	Purchase Cost	Day Rental Rate	Days Rented	Total Cost
CC1100 x 3	\$300.00	\$0.60	60	\$36.00
Development Board	?		60	
Experimenters Board	\$100.00	\$0.20	60	\$12.00
Oscope	\$8,000.00	\$16.00	60	\$960.00
			Total Rental Costs	\$1,008.00

Total Estimated Budget	\$23,058.00
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Budget: Week 15

Weekly Budget		(4/24/2008)		
Labor Costs				
Name	Hourly Pay Rate	Hours Worked	Total Pay	
Ronjan Mathur	\$15.00	150	\$2,250.00	
Aaron Adcock	\$15.00	150	\$2,250.00	
Will Gray	\$15.00	150	\$2,250.00	
Kyle Romero	\$15.00	150	\$2,250.00	
		Total Labor Cost	\$9,000.00	
		75% Overhead	\$6,750.00	
		Total Labor and Overhead	\$15,750.00	
Part Name	Cost	Quantity	Total Cost	
Board Printing	\$33.00	3	\$99.00	
Transmitter Board	\$15.77	2	\$31.54	
Receiver Board	\$22.79	1	\$22.79	
General Board Parts	\$151.82	1	\$151.82	
		Total Parts Cost	\$305.15	
Equipment Rental Costs				
Equipment Name	Purchase Cost	Day Rental Rate	Days Rented	Total Cost
CC1100 x 3	\$300.00	\$0.60	98	\$58.80
Experimenters Board	\$100.00	\$0.20	98	\$19.60
Oscope	\$8,000.00	\$16.00	98	\$1,568.00
			Total Rental Costs	\$1,646.40
Budget Total		\$17,701.55		

Gantt Chart

ID		Task Name	Start	Finish	Duration	% Complete	Resource Names
1		 Week 1	01/10/08	01/16/08	5d?	100	
2		Gain understanding of the CC1100	01/10/08	01/16/08	5d?	100	Mathur
3		Research into Packet/Modulation Techniques	01/10/08	01/16/08	5d?	100	Adcock
4		Look into the Programming Suite (SmartRF)	01/10/08	01/16/08	5d?	100	Romero
5		Began work with the Experimenter Board	01/10/08	01/16/08	5d?	100	Gray
6		Researched FCC Bands	01/10/08	01/10/08	1d?	100	Gray
7		 Week 2	01/17/08	01/23/08	5d?	0	
8		Attempt SmartRF communication with Experi...	01/17/08	01/23/08	5d?	0	Romero
9		Research Packet Creation	01/17/08	01/23/08	5d?	0	Adcock
10		Work on Transmitter Board	01/17/08	01/23/08	5d?	0	Gray
11		Review Sensor Setups	01/17/08	01/23/08	5d?	0	Mathur
12		 Week 3	01/24/08	01/30/08	5d?	0	
13		Look into CC1100 Registers / MSP430	01/24/08	01/30/08	5d?	0	Romero
14		Begin Creating Custom Packets	01/24/08	01/30/08	5d?	0	Adcock
15		Begin Transmitting with Transmitter Board	01/24/08	01/30/08	5d?	0	Gray
16		Order Sensors	01/24/08	01/30/08	5d?	0	Mathur
17		 Week 4	01/31/08	02/06/08	5d?	0	
18		Begin Timing Algorithm	01/31/08	02/06/08	5d?	0	Romero
19		Continue Working with Packets	01/31/08	02/06/08	5d?	0	Adcock
20		Begin working with MSP/Sensor Interface	01/31/08	02/06/08	5d?	0	Gray
21		Testing of Sensors	01/31/08	02/06/08	5d?	0	Mathur

Gantt Chart

ID		Task Name	Start	Finish	Duration	% Complete	Resource Names
22		 Week 5	02/07/08	02/13/08	5d?	0	
23		Research Sensor ID # Techniques	02/07/08	02/13/08	5d?	0	Romero
24		Testing Sensors with Transmission	02/07/08	02/13/08	5d?	0	Mathur
25		Transfer Sensor Data to packets	02/07/08	02/13/08	5d?	0	Gray
26		Finalize Packet Setup	02/07/08	02/13/08	5d?	0	Adcock
27		 Week 6	02/14/08	02/20/08	5d	0	
28		Begin testing packet Data	02/14/08	02/20/08	5d	0	Adcock
29		Begin Building Sensor/Transmitter Package	02/14/08	02/20/08	5d	0	Mathur
30		Begin Interpreting Packet Data	02/14/08	02/20/08	5d	0	Romero
31		Begin Power algorithm	02/14/08	02/20/08	5d	0	Gray
32		 Week 7	02/21/08	02/27/08	5d	0	
33		Begin Modulation Tests	02/21/08	02/27/08	5d	0	Adcock
34		Continue Building Sensor/Transmitter Package	02/21/08	02/27/08	5d	0	Mathur
35		Continue interpreting packet data	02/21/08	02/27/08	5d	0	Romero
36		Continue Power Algorithm	02/21/08	02/27/08	5d	0	Gray
37		 Week 8	02/28/08	03/05/08	5d	0	
38		Decide Modulation method	02/28/08	03/05/08	5d	0	Adcock
39		Finish Sensor/Transmitter Package	02/28/08	03/05/08	5d	0	Mathur
40		Finish Timing Algorithm	02/28/08	03/05/08	5d	0	Romero
41		Finish Power Algorithm	02/28/08	03/05/08	5d	0	Gray

Gantt Chart

42		Week 9	03/06/08	03/12/08	5d	0	
43		Test Packet Transmission (Separate)	03/06/08	03/12/08	5d	0	Adcock
44		Test Power Algorithm (Separate)	03/06/08	03/12/08	5d	0	Gray
45		Test Timing Algorithm (Separate)	03/06/08	03/12/08	5d	0	Romero
46		Design Full System Test	03/06/08	03/12/08	5d	0	Mathur
47		Week 10	03/13/08	03/26/08	10d	0	
48		Test Packet Transmission (Full System)	03/20/08	03/26/08	5d	0	Adcock
49		Test Power Algorithm (Full System)	03/20/08	03/26/08	5d	0	Gray
50		Test Timing/Receiver Algorithms (Full System)	03/20/08	03/26/08	5d	0	Romero
51		Test Sensors (Full System)	03/20/08	03/26/08	5d	0	Mathur
52		Week 11	03/13/08	03/19/08	5d	0	
53		Spring Break!	03/13/08	03/19/08	5d	0	All
54		Week 12	03/27/08	04/02/08	5d	0	
55		Fix Packet Problems	03/27/08	04/02/08	5d	0	
56		Fix Power Problems	03/27/08	04/02/08	5d	0	
57		Fix Receiver Problems	03/27/08	04/02/08	5d	0	
58		Fix Sensor Problems	03/27/08	04/02/08	5d	0	

Gantt Chart

59		Week 13	04/03/08	04/09/08	5d	0	
60		Continue System Testing	04/03/08	04/09/08	5d	0	Adcock
61		Continue System Testing	04/03/08	04/09/08	5d	0	Gray
62		Continue System Testing	04/03/08	04/09/08	5d	0	Romero
63		Continue System Testing	04/03/08	04/09/08	5d	0	Mathur
64		Week 14	04/10/08	04/16/08	5d	0	
65		Prepare Demonstration	04/10/08	04/16/08	5d	0	Romero
66		Make Poster For Demonstration	04/10/08	04/16/08	5d	0	Gray
67		Work out transmission errors	04/10/08	04/16/08	5d	0	Adcock
68		Work out Mechanical errors	04/10/08	04/16/08	5d	0	Mathur
69		Week 15	04/17/08	04/23/08	5d	0	
70		Demo Day!	04/17/08	04/23/08	5d	0	All
71		Week 16	04/24/08	04/30/08	5d	0	
72		Final Presentation/Report	04/24/08	04/30/08	5d	0	All

Mathur

Questions?

